JANUS : A Constraint-Aware Hybrid Photonic Architecture with One-Hot Optical RNS Multiplication, Distributed GaP Amplification, and Bounded High-Precision Computation

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***Abstract*—Large-scale photonic accelerators are fundamentally constrained by analog precision collapse, optical loss accumulation, thermal drift, nonlinear wavelength interference, and the unsustainable static power requirements of thermo-optic routing meshes. Conventional photonic matrix multiplication architectures rely upon analog amplitude accumulation followed by high-speed analog-to-digital conversion, implicitly assuming that signal fidelity can scale with matrix dimension. However, for a 128×128 photonic multiply-accumulate operation using 8-bit operands, the accumulated output spans more than 8.3×10^6 discrete levels, requiring approximately 138 dB signal-to-noise ratio (SNR) for lossless analog recovery—a precision-throughput tradeoff that becomes increasingly impractical under realistic shot noise, thermal noise, and ADC bandwidth constraints at multi-GHz operating frequencies. Mitigation techniques exist, but each substantially reduces effective throughput or efficiency.**

JANUS resolves this contradiction by abandoning analog amplitude reconstruction entirely. Instead of encoding numerical magnitude in optical intensity, JANUS reformulates computation as a spatial localization problem using One-Hot Optical Residue Number System (RNS) routing [1]–[3]. Each computation is represented as the position of light within 256 spatial waveguides, reducing photodetection to a binary decision: whether light exists within a single channel. This substantially reduces the required detection precision from a 138 dB analog precision target toward a binary threshold detection problem: whether signal power exceeds noise power in a single spatial channel. Binary threshold detection requires substantially lower detector precision and dynamic range than high-resolution analog reconstruction, while preserving exact integer residue computation.

The architecture further integrates a vertically pumped Gallium Phosphide (GaP) degenerate phase-sensitive amplification (PSA) framework to compensate routing loss without introducing substantial amplified spontaneous emission (ASE) noise [4], [5]. A non-volatile dilated Beneš routing fabric constructed from Phase-Change Material (PCM) switches substantially reduces the static power scaling of thermo-optic Mach-Zehnder Interferometer (MZI) meshes [6], [7]. Single-wavelength coherent fanout substantially suppresses dominant inter-channel nonlinear interference mechanisms including four-wave mixing (FWM) and cross-phase modulation (XPM) by architectural construction rather than post-hoc filtering.

To scale precision beyond native 8-bit operation, JANUS employs Spatial Residue Number System decomposition and exact digital reconstruction using the Chinese Remainder Theorem (CRT) [8]–[10]. This decouples arithmetic precision from analog optical fidelity. The resulting architecture operates entirely within bounded thermal, optical, and electrical regimes while sustaining deterministic exact integer computation.

Across three deployment configurations — Mini, Edge, and Datacenter — JANUS achieves sustained exact 64-bit throughput ranging from 57.6 TMAC/s to 919.2 TMAC/s under bounded operating conditions without continuous optical calibration. The architecture demonstrates that large-scale photonic AI acceleration becomes physically realizable only when analog precision recovery is replaced by spatially encoded exact arithmetic.

***Index Terms*—Photonic Computing, Optical Neural Networks, Residue Number Systems, One-Hot Encoding, Phase-Change Materials, Gallium Phosphide, Photonic AI Accelerators, Phase-Sensitive Amplification, Silicon Photonics, Thermal Management**

I. Introduction

The rapid scaling of artificial intelligence workloads has intensified interest in photonic computing architectures capable of exceeding the bandwidth-density and energy-efficiency limits of conventional electronic accelerators [11], [12]. Optical systems naturally provide massive parallelism, low propagation delay, and wavelength-domain multiplexing, making silicon photonics an attractive substrate for large-scale matrix multiplication and neural network acceleration [13]–[16].

Most contemporary photonic neural accelerators are constructed around Mach-Zehnder Interferometer (MZI) meshes performing analog optical multiply-accumulate operations [13], [17]–[20]. In these systems, numerical values are encoded into optical amplitudes, propagated through programmable interferometric meshes, and accumulated as continuous analog optical power before photodetection and analog-to-digital conversion.

While this paradigm demonstrates promising low-bit inference capability, it encounters severe scaling contradictions when extended toward large-scale exact computation. The contradiction originates from a fundamental mismatch between analog optical signal fidelity and the information density required by high-dimensional arithmetic accumulation.

Consider a conventional optical matrix multiplication involving 128 independent 8-bit activations and 128 independent 8-bit weights. The maximum unreduced analog accumulation is

$$S_{\max} = 128 \times 255 \times 255 \tag{1}$$

$$= 8{,}323{,}200 \tag{2}$$

Exact recovery of this value through analog photodetection requires resolution across more than 8.3×10^6 discrete analog levels. Under standard quantization theory, the minimum required SNR becomes [21], [22]

$$\mathrm{SNR}_{req} = 20 \log_{10} (8{,}323{,}200) \tag{3}$$
$$\approx 138.4 \text{ dB} \tag{4}$$

This requirement exceeds the practical operating range of modern photonic systems by a substantial margin. Shot noise, thermal noise, amplifier noise, detector nonlinearity, and finite ADC precision collectively prevent stable analog recovery at such information densities.

The failure is not merely an implementation limitation. It is a structural consequence of attempting to scale analog amplitude accumulation beyond physically sustainable entropy density.

Several mitigation strategies have been explored within prior work, including signal normalization, temporal bit slicing, calibration loops, and multi-stage electronic correction [23], [24]. However, these approaches only redistribute the precision burden rather than eliminating it. Temporal decomposition reduces throughput proportionally to bit depth, while aggressive normalization sacrifices dynamic range without solving the underlying Shannon-capacity limitation. Consequently, existing photonic accelerators remain constrained to low-bit approximate inference regimes.

JANUS originates from the observation that the analog accumulation model itself constitutes the primary scaling failure. Rather than attempting to improve analog precision recovery, JANUS reformulates optical computation into a spatial localization problem.

The central architectural transition is therefore:

$$\text{Amplitude Reconstruction} \quad \Longrightarrow \quad \text{Spatial Localization} \tag{5}$$

Instead of representing values through optical intensity, JANUS represents values through spatial waveguide position using One-Hot Optical Residue Number System (RNS) encoding. Each input activates exactly one waveguide among 256 possible spatial channels. Photodetection no longer estimates analog amplitude. Instead, the detector answers a binary question:

$$\text{“\textit{Does waveguide } } k \text{ \textit{contain light?}”} \tag{6}$$

This transforms the photodetection problem from high-resolution analog conversion into binary spatial detection. Binary threshold detection requires substantially lower detector precision and dynamic range than high-resolution analog reconstruction. The required detector condition therefore reduces to verifying that signal power exceeds the noise floor within the active channel:

$$\mathrm{SNR}_{binary} > 0 \text{ dB} \tag{7}$$

By substantially reducing the analog precision requirement, JANUS suppresses the dominant precision bottleneck preventing large-scale photonic exact arithmetic.

However, resolving the precision contradiction alone does not produce a physically realizable architecture. Large-scale photonic systems simultaneously encounter several additional scaling failures:

- Thermo-optic MZI routing meshes consume static hold power proportional to switch count, producing unsustainable thermal densities at scale.
- Dense wavelength-division multiplexing (DWDM) architectures suffer from nonlinear interference mechanisms including four-wave mixing (FWM) and cross-phase modulation (XPM).
- Long propagation paths accumulate optical routing loss faster than conventional amplifiers can compensate without introducing amplified spontaneous emission (ASE) corruption.
- Thermal diffusion from electronic subsystems destabilizes photonic phase coherence and induces calibration drift.

JANUS addresses these constraints through a fully constraint-driven architecture composed of five coupled mechanisms:

1) One-Hot Optical RNS multiplication for exact low-SNR binary detection.
2) Dilated non-volatile PCM Beneš routing networks substantially reducing static hold power.
3) Single-wavelength coherent spatial fanout removing inter-channel nonlinear interference.
4) Distributed vertically pumped GaP degenerate phase-sensitive amplification compensating routing loss with substantially reduced noise accumulation under phase-matched conditions.
5) Spatial CRT reconstruction and JIR-predictive RRNS fault management enabling exact precision scaling with deterministic error detectability.

The resulting system operates entirely within bounded thermal, optical, and electrical regimes:

$$P_{noise} < P_{signal} < P_{sat} \tag{8}$$

$$T_{junction} < T_{critical} \tag{9}$$

$$\phi_{drift} < \phi_{max} \tag{10}$$

Rather than pursuing theoretical peak throughput under idealized assumptions, JANUS prioritizes sustained deterministic operation under realistic deployment conditions.

II. Physical Constraint Landscape

A. Analog Precision Collapse in Photonic Accumulation

The dominant assumption underlying conventional optical neural networks is that analog optical accumulation can scale

TABLE I
ANALOG ACCUMULATION SCALING CONSTRAINTS ACROSS JANUS CONFIGURATIONS

Metric	Mini	Edge	Datacenter
Mesh Size	32×32	64×64	128×128
Maximum Accumulation	2,080,800	4,161,600	8,323,200
Required Analog SNR	126.4 dB	132.4 dB	138.4 dB
ADC Feasibility at 33 GHz	Increasingly impractical	Increasingly impractical	Increasingly impractical
Operational Stability	Unbounded	Unbounded	Unbounded

proportionally with matrix dimension while maintaining recoverable signal fidelity. This assumption fails under realistic large-scale operating conditions.

For an $N \times N$ photonic matrix multiplication using unsigned 8-bit operands, the worst-case analog accumulation scales as

$$S_{\max} = N \times 255 \times 255 \tag{11}$$

For the three JANUS deployment configurations:

$$\begin{aligned} S_{\max,\mathrm{Mini}} &= 32 \times 255 \times 255 && (12)\\ &= 2{,}080{,}800 && (13)\end{aligned}$$

$$\begin{aligned} S_{\max,\mathrm{Edge}} &= 64 \times 255 \times 255 && (14)\\ &= 4{,}161{,}600 && (15)\end{aligned}$$

$$\begin{aligned} S_{\max,\mathrm{Datacenter}} &= 128 \times 255 \times 255 && (16)\\ &= 8{,}323{,}200 && (17)\end{aligned}$$

The corresponding minimum analog SNR requirements become

$$\mathrm{SNR}_{req} = 20 \log_{10} (S_{\max}) \tag{18}$$

Thus:

$$\begin{aligned} \mathrm{SNR}_{Mini} &= 20 \log_{10}(2{,}080{,}800) && (19)\\ &\approx 126.4 \text{ dB} && (20)\end{aligned}$$

$$\begin{aligned} \mathrm{SNR}_{Edge} &= 20 \log_{10}(4{,}161{,}600) && (21)\\ &\approx 132.4 \text{ dB} && (22)\end{aligned}$$

$$\begin{aligned} \mathrm{SNR}_{Datacenter} &= 20 \log_{10}(8{,}323{,}200) && (23)\\ &\approx 138.4 \text{ dB} && (24)\end{aligned}$$

These requirements exceed realistic photonic operating conditions by orders of magnitude.

The contradiction becomes especially severe when detector bandwidth is considered. At the target JANUS operating frequency of approximately 33 GHz, no practical ADC architecture can simultaneously sustain high resolution, high sampling rate, low thermal noise, and acceptable power density within a realistic operating envelope.

Consequently, the analog accumulation paradigm itself becomes the primary scaling failure.

B. Failure of Conventional Precision Recovery Techniques

Several conventional mitigation strategies attempt to preserve analog photonic scaling.

Voltage normalization compresses signal amplitude but does not reduce information density. The required recoverable entropy remains unchanged, meaning the ADC precision burden persists despite lower voltage swing.

Temporal bit slicing distributes precision across multiple cycles:

$$T_{effective} = \frac{T_{native}}{B} \tag{25}$$

where B is the number of temporal slices.

For 16-cycle decomposition:

$$T_{effective} = \frac{33 \text{ GHz}}{16} \approx 2.06 \text{ GHz} \tag{26}$$

This destroys the throughput advantage motivating photonic acceleration in the first place.

Similarly, iterative calibration loops attempt to compensate thermal drift and phase instability through repeated electronic correction. However, calibration complexity scales with:

$$\mathcal{O}(N^2) \tag{27}$$

for dense interferometric meshes, making large-scale stable operation thermally impractical.

Therefore, existing mitigation strategies merely redistribute the precision failure rather than resolving it fundamentally.

C. Static Power Explosion in Thermo-Optic Routing Meshes

Large-scale silicon photonic accelerators commonly employ thermo-optic Mach-Zehnder Interferometer (MZI) meshes for programmable optical routing and weighting [12], [17]. In such systems, optical phase is controlled through resistive heating, which continuously modifies the refractive index of silicon via the thermo-optic effect.

Although effective for small-scale interferometric control, thermo-optic routing exhibits catastrophic static power scaling at large switch counts.

For a conventional thermo-optic MZI switch, the static hold power is typically:

$$P_{\mathrm{MZI}} \approx 10 \text{ mW} \tag{28}$$

The total routing hold power therefore scales linearly with the number of switches:

TABLE II
THERMO-OPTIC ROUTING POWER SCALING FAILURE

Metric	Mini	Edge	Datacenter
Switch Count	62,914,560	251,658,240	1,006,632,960
Per-Switch Hold Power	10 mW	10 mW	10 mW
Total Static Hold Power	629.14 kW	2,516.58 kW	10,066.32 kW
Thermal Feasibility	Increasingly impractical	Increasingly impractical	Increasingly impractical
Continuous Bias Requirement	Yes	Yes	Yes

$$P_{\text{routing}} = N_{\text{switch}} \times P_{\text{MZI}} \tag{29}$$

The total PCM Beneš switch count for the three JANUS configurations represents the complete system-total across all multipliers and tiles. Each per-multiplier Beneš fabric instance contains 1,920 switches (256-port, 15 stages); the figures below scale this by all N^2 multipliers per tile and the total number of tiles:

$$N_{\text{Mini}} = 32 \times 32^2 \times 1,920 = 62,914,560 \tag{30}$$

$$N_{\text{Edge}} = 32 \times 64^2 \times 1,920 = 251,658,240 \tag{31}$$

$$N_{\text{Datacenter}} = 32 \times 128^2 \times 1,920 = 1,006,632,960 \tag{32}$$

The complete Datacenter system therefore contains approximately 1.006 billion PCM switches across 524,288 per-multiplier Beneš fabrics. (See Appendix B for the single-fabric derivation.)

If implemented using conventional thermo-optic MZI heaters, the corresponding static hold powers become:

$$P_{\text{Mini}} = 62{,}914{,}560 \times 10 \text{ mW} \tag{33}$$

$$= 629.14 \text{ kW} \tag{34}$$

$$P_{\text{Edge}} = 251{,}658{,}240 \times 10 \text{ mW} \tag{35}$$

$$= 2,516.58 \text{ kW} \tag{36}$$

$$P_{\text{Datacenter}} = 1{,}006{,}632{,}960 \times 10 \text{ mW} \tag{37}$$

$$= 10,066.32 \text{ kW} \tag{38}$$

These values exceed practical chip-scale thermal extraction capability by several orders of magnitude.

The failure mechanism is fundamentally thermodynamic. The generated heat modifies the local refractive index of adjacent waveguides:

$$\Delta n = \frac{dn}{dT}\Delta T \tag{39}$$

where:

$$\frac{dn}{dT} \approx 1.86 \times 10^{-4} \text{ K}^{-1} \tag{40}$$

for silicon photonics [22], [25].

As routing density increases, lateral heat diffusion produces cross-coupled phase drift across neighboring interferometers. The resulting thermal interference destabilizes the optical transfer matrix and forces continuous recalibration.

Thus, the routing fabric itself becomes the dominant thermal failure mechanism in large-scale photonic systems.

D. Nonlinear Failure of Multi-Wavelength Scaling

Many photonic accelerator architectures attempt to improve throughput through Dense Wavelength-Division Multiplexing (DWDM). In these systems, multiple optical carrier frequencies propagate simultaneously through shared waveguides.

While spectrally multiplexed operation increases theoretical bandwidth, it introduces nonlinear optical interference mechanisms that scale superlinearly with optical power density.

The dominant nonlinear effects are:

- Four-Wave Mixing (FWM)
- Cross-Phase Modulation (XPM)
- Self-Phase Modulation (SPM)
- Spectral Beating Noise

In conventional DWDM systems, the nonlinear interference noise power scales approximately as [26], [27]:

$$\sigma^2_{\text{NL}} \ \propto \ \gamma^2 P^3 L^2_{\text{eff}}/A_{\text{eff}}, \tag{41}$$

where γ is the nonlinear coefficient, P is the optical power, L_{eff} is the effective interaction length, and A_{eff} is the effective mode area. The quadratic dependence on γ and L_{eff} captures the higher-order nature of four-wave mixing, while the P^3 term reflects the cubic growth with launch power.

As channel density increases, multiple wavelength combinations satisfy phase-matching conditions, producing ghost frequencies through inter-channel mixing.

The resulting spectral contamination corrupts the recoverability of analog optical information.

Importantly, this failure cannot be completely removed through filtering. Filtering suppresses generated sidebands only after nonlinear mixing has already occurred. Therefore, the underlying epistemic corruption of the optical signal persists.

The problem becomes increasingly severe under high-power amplification, where stronger nonlinear interaction lengths accelerate spectral mixing.

Consequently, wavelength scaling introduces a contradiction:

$$\text{Higher Spectral Density} \implies \text{Higher Nonlinear Corruption} \tag{42}$$

This makes large-scale exact arithmetic fundamentally unstable under multi-wavelength analog operation.

E. Optical Loss Accumulation and Amplifier Noise

Large photonic routing fabrics inevitably accumulate substantial optical propagation loss. The total path loss can be expressed as:

$$L_{\text{total}} = L_{\text{mod}} + L_{\text{wg}} + L_{\text{coupler}} + L_{\text{MZI}} + L_{\text{routing}} \quad (43)$$

For practical large-scale systems:

$$L_{\text{mod}} \approx 3\text{–}6 \text{ dB} \quad (44)$$

$$L_{\text{wg}} \approx 0.5\text{–}2 \text{ dB} \quad (45)$$

$$L_{\text{coupler}} \approx 1\text{–}3 \text{ dB} \quad (46)$$

$$L_{\text{routing}} \approx 15\text{–}20 \text{ dB} \quad (47)$$

Thus:

$$L_{\text{total}} \approx 30\text{–}50 \text{ dB} \quad (48)$$

Without compensation, optical power decays exponentially along the propagation path:

$$P(z) = P_0 e^{-\alpha z} \quad (49)$$

where: α is the effective attenuation coefficient.

Conventional semiconductor optical amplifiers (SOAs) attempt to restore signal amplitude through stimulated emission. However, SOAs introduce Amplified Spontaneous Emission (ASE) noise [21], [22]:

$$\sigma^2_{\text{ASE}} \propto G n_{sp} B \quad (50)$$

where: G is amplifier gain, n_{sp} is the spontaneous emission factor, and: B is bandwidth.

The contradiction is therefore:

$$\text{Higher Gain} \implies \text{Higher ASE Corruption} \quad (51)$$

Thus, conventional amplification preserves optical amplitude while simultaneously degrading informational integrity.

For exact arithmetic systems, this failure is especially severe because even small analog corruption propagates catastrophically through high-dimensional reconstruction.

F. Thermal Drift and Calibration Instability

Photonic interferometric systems are intrinsically sensitive to thermal variation because optical phase depends directly upon refractive index and propagation length.

The accumulated optical phase is:

$$\phi = \frac{2\pi n L}{\lambda} \quad (52)$$

Thermal perturbation therefore produces phase drift:

$$\Delta\phi = \frac{2\pi L}{\lambda}\frac{dn}{dT}\Delta T \quad (53)$$

As routing density increases, neighboring thermal sources generate dynamic phase instability throughout the photonic mesh.

The resulting transfer matrix becomes time-varying:

$$H(t) \neq H(t + \Delta t) \quad (54)$$

Conventional architectures compensate this instability through active calibration loops involving:

- thermal tuning,
- phase locking,
- electronic correction,
- iterative optimization,
- feedback control.

However, calibration overhead scales poorly with mesh density [12], [23]:

$$C_{\text{calibration}} \propto N^2 \quad (55)$$

At large dimensions, calibration itself becomes a dominant energy and latency cost.

Furthermore, continuously recalibrated systems do not operate within a stable thermodynamic equilibrium. Instead, they remain permanently trapped in transient correction states.

Thus, large-scale photonic accelerators face a second-order contradiction:

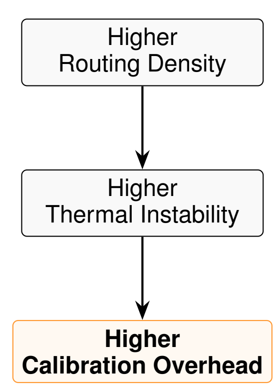


Fig. 1. Causal impact of physical routing constraints on operational runtime overhead.

This contradiction prevents sustained deterministic operation under realistic deployment conditions.

G. Constraint-Driven Architectural Reformulation

The preceding constraints collectively imply that large-scale photonic exact arithmetic cannot emerge from incremental improvement of conventional analog architectures.

The failure mechanisms are structural rather than implementation-level.

Specifically:

- Analog accumulation fails due to Shannon-limited precision collapse.
- Thermo-optic routing fails due to unsustainable static power density.
- Multi-wavelength scaling fails due to nonlinear spectral corruption.
- Conventional amplification fails due to ASE-induced epistemic loss.
- Dense interferometric meshes fail due to thermal calibration instability.

Consequently, JANUS adopts a fundamentally different computational philosophy.

Rather than attempting to improve analog reconstruction fidelity, JANUS reformulates optical computation into a bounded spatial routing problem.

The resulting architecture enforces:

$$P_{\text{noise}} < P_{\text{signal}} < P_{\text{sat}} \tag{56}$$

$$T_{\text{junction}} < T_{\text{critical}} \tag{57}$$

$$\phi_{\text{drift}} < \phi_{\max} \tag{58}$$

within all operating regimes.

This bounded-operation methodology intentionally sacrifices theoretical peak analog density in favor of sustained exact arithmetic under realistic physical constraints.

III. Constraint-Driven Architectural Response

A. Architectural Overview

JANUS resolves the scaling contradictions identified in the previous section through a tightly coupled set of architectural mechanisms, each directly derived from a specific physical failure mode.

The resulting system is not an incremental extension of conventional analog photonic computing. Instead, it is a reformulation of photonic computation around bounded spatial exact arithmetic.

The architecture consists of six tightly integrated subsystems:

1) One-Hot Optical Residue Number System multiplication,
2) Dilated PCM Beneš routing fabrics,
3) Distributed vertically pumped GaP PSA amplification,
4) Single-wavelength coherent spatial fanout,
5) Spatial CRT precision reconstruction,
6) Thermally decoupled 3D Z-axis integration.

Each subsystem exists because a conventional alternative becomes physically unstable at large scale.

The resulting architecture operates under a constraint-first design principle:

$$\text{Physical Stability} > \text{Theoretical Peak Density} \tag{59}$$

Unlike conventional photonic accelerators that rely upon continuous calibration and optimistic analog assumptions, JANUS constrains all thermal, optical, and electrical states within bounded operating regions.

This bounded-operation philosophy forms the foundational principle of the entire architecture.

a) Demonstrated Components and Forward-Looking Integration.: Several device-level components discussed in this work have been independently demonstrated in literature, including high-bandwidth germanium photodiodes (up to 265 GHz), GaP nonlinear optical elements, $Ge_2Sb_2Te_5$ PCM switching in silicon photonic waveguides, and silicon photonic routing meshes. However, the full JANUS architecture as described—including the distributed PSA fabric, multi-million-switch non-volatile Beneš fabric, complete stacked integration, and multi-tile datacenter configuration—remains a forward-looking systems integration proposal. This paper presents the architectural and physical motivation for such integration rather than a fabrication-validated implementation.

B. One-Hot Optical Residue Number System Multiplication

The central architectural transition within JANUS is the elimination of analog optical accumulation. Instead of encoding numerical information into continuously varying optical amplitudes, JANUS reformulates arithmetic as a spatial routing problem using One-Hot Optical Residue Number System (RNS) encoding [1]–[3].

This transition is motivated by the observation that analog amplitude reconstruction constitutes the dominant precision bottleneck in large-scale photonic computation.

1) Failure of Analog Optical Accumulation: Conventional optical neural accelerators encode values through optical power:

$$x_i \rightarrow P_i \tag{60}$$

Weighted optical contributions are then accumulated through coherent or incoherent interference:

$$P_{\text{out}} = \sum_{i=1}^{N} w_i x_i \tag{61}$$

The photodetector must subsequently reconstruct the analog output value through high-resolution amplitude estimation.

For an unsigned 8-bit accumulation:

$$x_i, w_i \in [0, 255] \tag{62}$$

the maximum accumulation becomes:

$$S_{\max} = N \times 255 \times 255 \tag{63}$$

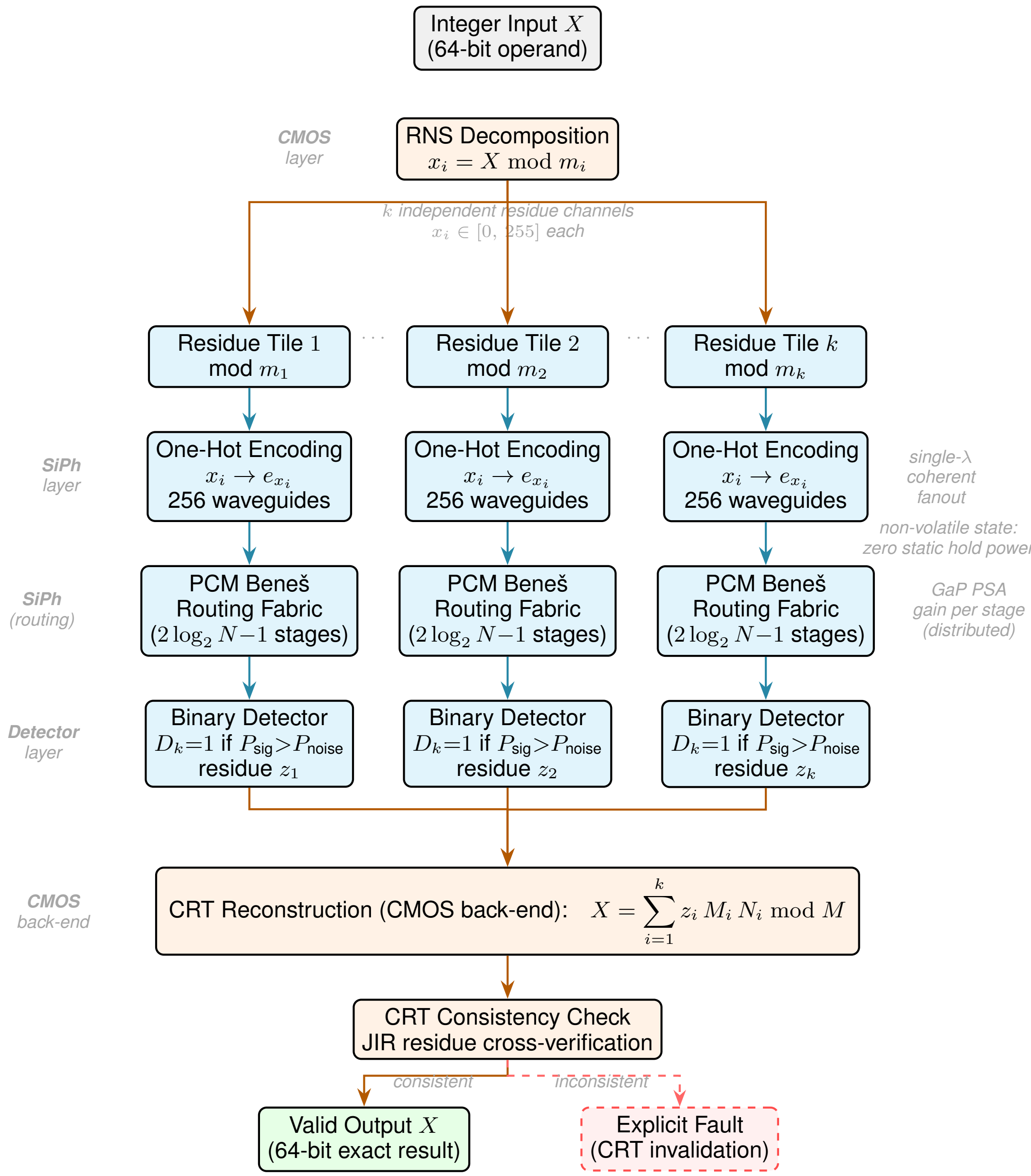


Fig. 2. JANUS end-to-end data and computation flow. A 64-bit input integer is decomposed into k independent residues by the CMOS front-end (RNS Decomposition). Each residue is encoded spatially as a one-hot vector across 256 waveguides within its dedicated optical tile. A coherent single-wavelength fanout distributes light; a non-volatile dilated PCM Beneš fabric performs deterministic spatial permutation corresponding to modular multiplication, with distributed GaP PSA compensating per-stage routing loss. Binary photodetectors read the active waveguide position, producing the modular product z_i. The CMOS back-end reconstructs the exact 64-bit result via the Chinese Remainder Theorem and applies the JIR consistency check: a coherent residue set yields a valid output, while any CRT inconsistency triggers an explicit fault rather than a silent incorrect result.

As previously derived:

$$S_{\text{max,Mini}} = 2{,}080{,}800 \tag{64}$$

$$S_{\text{max,Edge}} = 4{,}161{,}600 \tag{65}$$

$$S_{\text{max,Datacenter}} = 8{,}323{,}200 \tag{66}$$

The corresponding minimum exact analog SNR requirements are:

$$\text{SNR}_{\text{Mini}} \approx 126.4 \text{ dB} \tag{67}$$

$$\text{SNR}_{\text{Edge}} \approx 132.4 \text{ dB} \tag{68}$$

$$\text{SNR}_{\text{Datacenter}} \approx 138.4 \text{ dB} \tag{69}$$

These requirements exceed realistic photonic operating conditions.

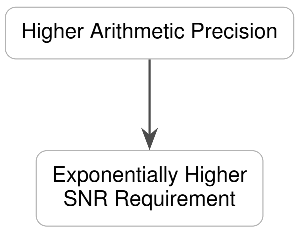
The failure mechanism is fundamentally informational. The detector must estimate not merely the presence of light, but its precise analog magnitude across millions of distinguishable states.

Under Shannon channel theory, recoverable information scales as [21], [22]:

$$C = B \log_2 (1 + \mathrm{SNR}) \tag{70}$$

Thus, increasing analog precision directly demands exponentially larger SNR.

This produces the core contradiction of analog photonic computation:



At practical operating frequencies near 33 GHz, no physically realizable photonic detector chain can satisfy these requirements while maintaining acceptable thermal density and power consumption [28]–[30].

2) Transition from Amplitude Reconstruction to Spatial Localization: JANUS resolves this contradiction by abandoning analog amplitude representation entirely.

Instead of encoding values through optical intensity, JANUS encodes values spatially.

Each residue value activates exactly one optical waveguide among 256 possible channels:

$$x \rightarrow e_x \tag{71}$$

where:

$$e_x = (0, 0, \ldots, 1, \ldots, 0) \tag{72}$$

is a one-hot spatial basis vector.

Thus, arithmetic state is represented through waveguide position rather than analog optical amplitude.

Optical multiplication becomes deterministic spatial routing:

$$(a, b) \rightarrow (a \cdot b) \mod p \tag{73}$$

where: p is the local residue modulus.

The photodetector therefore no longer reconstructs analog magnitude. Instead, it performs binary localization:

$$D_k = \begin{cases} 1, & \text{light present in channel } k \\ 0, & \text{otherwise} \end{cases} \tag{74}$$

This transforms the computational problem from:

$$\text{Analog Magnitude Estimation} \tag{75}$$

into:

$$\text{Binary Spatial Localization} \tag{76}$$

The required detector condition therefore collapses to:

$$P_{\text{signal}} > P_{\text{noise}} \tag{77}$$

which corresponds approximately to:

$$\mathrm{SNR}_{\text{binary}} > 0 \text{ dB} \tag{78}$$

While the Ge-fin detector possesses a 265 GHz intrinsic bandwidth, the receiver circuit is low-pass filtered to match the 33 GHz system clock. This bandwidth reduction drops the effective thermal noise floor to approximately -31.76 dBm, providing a ≈12 dB binary detection margin for a -20 dBm target signal.

Thus, the architecture transforms a 138 dB analog precision requirement into a near-zero binary detection threshold.

TABLE III
TRANSITION FROM ANALOG ACCUMULATION TO ONE-HOT SPATIAL DETECTION

Metric	Mini	Edge	Datacenter
Maximum Analog Accumulation	2,080,800	4,161,600	8,323,200
Required Analog SNR	126.4 dB	132.4 dB	138.4 dB
One-Hot Detection Threshold	> -30 dBm	> -30 dBm	> -30 dBm
ADC Requirement	Eliminated	Eliminated	Eliminated
Detection Type	Binary	Binary	Binary
Operational Stability	Bounded	Bounded	Bounded

3) One-Hot Residue Arithmetic: JANUS performs arithmetic independently within residue domains.

For a modulus set:

$$\mathcal{M} = \{m_1, m_2, \ldots, m_k\} \tag{79}$$

a large integer: X is decomposed into residue components:

$$X \rightarrow (x_1, x_2, \ldots, x_k) \tag{80}$$

where:

$$x_i = X \mod m_i \tag{81}$$

Each residue channel operates independently:

$$z_i = (x_i y_i) \mod m_i \tag{82}$$

Importantly, no analog carry propagation exists between channels.

This eliminates the principal scaling bottleneck of conventional binary arithmetic [1], [8]:

$$\text{Carry Propagation Delay} \rightarrow 0 \tag{83}$$

because each residue tile computes independently.

The resulting architecture therefore scales spatially rather than temporally.

4) Optical Routing Interpretation of Multiplication: Within JANUS, multiplication is implemented physically as deterministic waveguide permutation.

For modulus:

$$m \tag{84}$$

the routing function becomes:

$$f(a,b) = (a \cdot b) \mod m \tag{85}$$

The optical mesh therefore behaves as a programmable permutation tensor.

Unlike analog MAC systems, no weighted accumulation occurs. Optical energy is conserved spatially:

$$\sum_i P_i = P_{\text{total}} \tag{86}$$

Thus, information encoding no longer depends upon precise analog optical linearity.

Instead, correctness depends solely upon successful spatial routing.

This distinction fundamentally changes the role of photonics within the computational stack.

Conventional systems attempt to use optics as analog arithmetic engines.

JANUS instead uses optics as:

$$\textit{high-bandwidth deterministic routing media} \tag{87}$$

while exact reconstruction is delegated to modular arithmetic and CMOS verification logic.

5) Detection Margin and False Positive Suppression: The effective signal margin within JANUS is determined by:

$$M = P_{\text{signal}} - P_{\text{noise}} \tag{88}$$

For the nominal routing path, the optical chain proceeds as follows: each external laser (peak power) is passively split $1:64$ (loss 18.06 dB) so that the power just before the GaP amplification layer is -20.0 dBm. The GaP parametric gain (via microring resonant cavity) is $+30$ dB, and the PCM Beneš routing network (15 stages $\times 0.5$ dB/stage) introduces -7.5 dB loss. For a conservative estimate, we assume a higher routing loss of -10.0 dB, accounting for potential additional insertion losses from waveguide crossings, fabrication variations, or thermal effects. Thus the signal power at the Ge-fin detector is approximately:

$$P_{\text{signal}} \approx -7.5 \text{ dBm}. \tag{89}$$

The Ge-fin detector dark noise floor, after low-pass filtering the TIA receiver to match the 33 GHz system clock (well below the 265 GHz intrinsic bandwidth), is:

$$P_{\text{noise floor}} \approx -31.76 \text{ dBm}. \tag{90}$$

Therefore the binary detection margin is:

$$M = (-7.5) - (-31.76) = +24.26 \text{ dB}. \tag{91}$$

This margin substantially exceeds the binary detection threshold.

Consequently, detector uncertainty does not accumulate analog precision error across routing depth.

Instead, the system remains within a stable binary operating regime:

$$P_{\text{noise}} < P_{\text{signal}} < P_{\text{sat}} \tag{92}$$

throughout the computational pipeline.

6) Architectural Consequences: The transition from analog accumulation to One-Hot spatial routing produces several system-level consequences.

- First, high-resolution ADCs are completely removed from the optical pipeline.
- Second, thermal power density collapses because detector complexity no longer scales with arithmetic precision.
- Third, arithmetic exactness becomes decoupled from analog optical fidelity.
- Fourth, optical routing depth no longer compounds analog quantization error.
- Finally, the photonic subsystem operates entirely within bounded binary decision regions rather than unstable high-resolution analog regimes.

Thus, JANUS does not improve analog photonic precision. It restructures the computational problem so that exact integer output no longer depends upon high-resolution analog amplitude recovery.

C. Dilated PCM Beneš Routing Fabric

After eliminating analog accumulation, the next dominant scaling constraint becomes optical routing density. Large-scale One-Hot spatial arithmetic requires millions of deterministic waveguide permutations operating simultaneously with bounded thermal and optical behavior.

Conventional thermo-optic interferometric meshes cannot satisfy these requirements because their static power consumption scales linearly with switch count and continuously perturbs neighboring photonic states.

JANUS therefore replaces actively biased thermo-optic routing with a non-volatile Phase-Change Material (PCM) Beneš routing fabric [6], [31], [32].

1) Failure of Thermo-Optic Routing at Scale: Conventional silicon photonic meshes rely upon thermo-optic Mach-Zehnder Interferometers (MZIs) for programmable switching. Each interferometer maintains routing state through continuous resistive heating.

The steady-state power per switch is approximately:

$$P_{\text{MZI}} \approx 10 \text{ mW} \tag{93}$$

For a routing network containing:

$$N_{\text{switch}} \tag{94}$$

switches, the total static hold power becomes:

$$P_{\text{hold}} = N_{\text{switch}} P_{\text{MZI}} \tag{95}$$

As derived in Section II.C, conventional thermo-optic routing would demand 314.57 kW, 2516.58 kW, and $10,066.32$ kW respectively for the Mini, Edge, and Datacenter configurations (Table II). These power densities exceed practical chip-scale thermal extraction capability.

Furthermore, the generated heat induces dynamic refractive-index drift:

$$\Delta n = \frac{dn}{dT} \Delta T \tag{96}$$

which subsequently perturbs optical phase:

$$\Delta\phi = \frac{2\pi L}{\lambda} \frac{dn}{dT} \Delta T \tag{97}$$

As routing density increases, neighboring interferometers thermally couple through lateral diffusion, destabilizing the transfer matrix of the optical mesh.

The architecture therefore enters a self-reinforcing instability cycle:

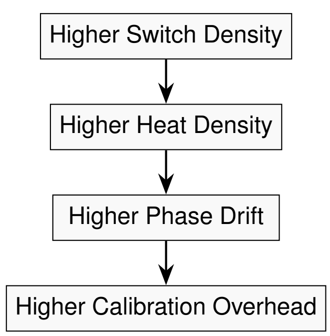


Fig. 3. Causal mechanism of thermal-induced phase drift in the proposed architecture.

Thus, thermo-optic routing fundamentally fails to scale toward large-scale exact photonic arithmetic.

2) PCM-Based Non-Volatile Optical Routing: JANUS resolves this contradiction through non-volatile plasmonic PCM routing elements constructed from:

$$\mathrm{Ge_2Sb_2Te_5} \tag{98}$$

(GST) [6], [33].

Unlike thermo-optic MZIs, PCM switches consume energy only during state transition [6], [7], [33].

The switching energy is:

$$E_{\text{PCM}} \approx 10\text{–}100 \text{ pJ} \tag{99}$$

After programming, the optical state persists without continuous power.

Thus:

$$P_{\text{hold}} \approx 0 \text{ W} \tag{100}$$

for all steady-state routing configurations.

This transforms routing power scaling from:

$$P \propto N_{\text{switch}} \tag{101}$$

into:

$$P \propto N_{\text{reconfiguration}} \tag{102}$$

where only actively reprogrammed switches consume energy.

Consequently, the continuous static thermal load of conventional MZI meshes is substantially reduced under infrequent-reconfiguration workloads.

TABLE IV
COMPARISON BETWEEN THERMO-OPTIC AND PCM ROUTING FABRICS

Metric	**Thermo-Optic MZI**	**JANUS PCM Fabric**
State Retention	Volatile	Non-Volatile
Static Hold Power	$\sim$ 10 mW/switch	$\approx$ 0 W
Thermal Drift	High	Minimal
Continuous Bias	Required	Not Required
Calibration Frequency	Continuous	Sparse
Thermal Stability	Unbounded	Bounded
Large-Scale Feasibility	No	Yes

3) Why Beneš Topology: After eliminating static hold power, the routing topology itself becomes critical.

JANUS adopts a Beneš network because it provides rearrangeable nonblocking routing with logarithmic stage scaling.

For an N-port Beneš network, the total stage count is:

$$S = 2\log_2(N) - 1 \tag{103}$$

The JANUS One-Hot encoding uses 256 spatial waveguides per multiplier input, regardless of the matrix dimension N. The Beneš routing granularity is therefore the **256-waveguide** One-Hot channel count, not the matrix dimension. Applying this universally:

$$S = 2\log_2(256) - 1 = 2 \times 8 - 1 = \mathbf{15} \tag{104}$$

Thus:[1]

$$S_{\text{Mini}} = S_{\text{Edge}} = S_{\text{Datacenter}} = 15 \tag{105}$$

The corresponding switch complexity scales as:

$$N_{\text{switch}} = \frac{N}{2}(2\log_2(N) - 1) \tag{106}$$

This logarithmic depth is substantially more scalable than fully connected crossbar routing.

A crossbar requires: N^2 switches, which becomes physically prohibitive at large dimensions.

[1]The stage count is identical across all three deployment tiers because the Beneš network operates on the fixed 256-waveguide One-Hot alphabet, not on the matrix dimension N.

TABLE V
BENEŠ ROUTING SCALING ACROSS JANUS CONFIGURATIONS

Metric	Mini	Edge	Datacenter
Mesh Dimension	32×32	64×64	128×128
One-Hot Waveguide Count	256	256	256
Beneš Stages (on 256-port)	15	15	15
Routing Complexity	$\mathcal{O}(N \log N)$	$\mathcal{O}(N \log N)$	$\mathcal{O}(N \log N)$
Crossbar Complexity	$\mathcal{O}(N^2)$	$\mathcal{O}(N^2)$	$\mathcal{O}(N^2)$
Nonblocking Routing	Yes	Yes	Yes

The Beneš structure therefore minimizes routing depth while preserving universal permutation capability.

4) Dilated Routing and Crosstalk Suppression: Conventional dense optical meshes suffer from waveguide crossings and switch contention. As optical density increases, parasitic coupling generates coherent crosstalk:

$$P_{\mathrm{XT}} \propto \kappa P_{\mathrm{signal}} \tag{107}$$

where: κ is the coupling coefficient.

Even weak parasitic coupling becomes problematic under exact arithmetic, because coherent interference introduces false optical activations.

JANUS suppresses this failure mode using a dilated Beneš topology.

In the dilated architecture, active and inactive routing paths are physically separated:

$$\text{Active Path} \cap \text{Dummy Path} = \varnothing \tag{108}$$

Thus, simultaneously routed optical channels never share identical switching elements.

This spatial separation suppresses coherent leakage accumulation and maintains binary detection integrity.

Importantly, crosstalk suppression is achieved structurally rather than through post-hoc correction.

The architecture therefore suppresses the interference mechanism at the structural level rather than attempting to compensate it afterward.

5) Routing Loss Characteristics: Each Beneš stage introduces finite insertion loss. The GST/PCM switch extinction ratio and insertion loss is taken as 0.5 dB per stage, consistent with demonstrated plasmonic-PCM hybrid switch platforms:

$$L_{\mathrm{stage}} \approx 0.5 \text{ dB} \tag{109}$$

Total routing loss therefore scales as:

$$L_{\mathrm{routing}} = S \times L_{\mathrm{stage}} \tag{110}$$

Since all tiers use 15 Beneš stages on the 256-waveguide One-Hot fabric:

$$L_{\mathrm{Mini}} = L_{\mathrm{Edge}} = L_{\mathrm{Datacenter}} = 15 \times 0.5 = 7.5 \text{ dB} \tag{111}$$

Although routing loss remains unavoidable, it stays within a bounded operating regime compatible with distributed optical amplification.

TABLE VI
ROUTING LOSS SCALING IN DILATED PCM BENEŠ NETWORKS

Metric	Mini	Edge	Datacenter
Beneš Stages (256-port universal)	15	15	15
Per-Stage Loss (plasmonic PCM)	0.5 dB	0.5 dB	0.5 dB
Total Routing Loss	7.5 dB	7.5 dB	7.5 dB
Static Hold Power	0 W	0 W	0 W
Thermal Stability	Bounded	Bounded	Bounded

6) Architectural Consequences: The PCM Beneš fabric fundamentally changes the scaling behavior of large photonic routing systems.

- First, static thermal power collapses from kilowatt-scale continuous dissipation to near-zero steady-state operation.
- Second, routing depth scales logarithmically rather than quadratically.
- Third, nonblocking permutation capability enables deterministic One-Hot residue routing without dynamic contention.
- Fourth, dilated path separation suppresses coherent crosstalk at the architectural level.
- Finally, the routing network becomes thermodynamically stable under sustained operation because routing state no longer requires continuous electrical bias.

Thus, JANUS does not optimize conventional thermo-optic meshes.

It substantially mitigates the thermodynamic failure mode that prevents their large-scale deployment.

D. Distributed Vertically Pumped GaP Phase-Sensitive Amplification

After eliminating analog accumulation and static routing power, the next dominant scaling constraint becomes optical propagation loss.

Even under low-loss silicon photonics, large Beneš routing fabrics accumulate several decibels of attenuation across long optical paths. Without amplification, signal power eventually falls below binary detection thresholds.

Conventional semiconductor optical amplifiers (SOAs) partially restore optical amplitude but introduce amplified spontaneous emission (ASE) noise that corrupts exact arithmetic recoverability.

JANUS therefore employs distributed vertically pumped Gallium Phosphide (GaP) degenerate phase-sensitive amplification (PSA) to compensate routing loss while preserving epistemic integrity of the optical signal.

1) Failure of Conventional Optical Amplification: Optical propagation loss accumulates continuously across routing stages, waveguide bends, couplers, and modulation interfaces.

The total optical power after propagation distance z is:

$$P(z) = P_0 e^{-\alpha z} \tag{112}$$

where: α is the effective attenuation coefficient.

As previously derived, routing loss within the JANUS Beneš network (15 stages at 0.5 dB/stage on the 256-port One-Hot fabric) is uniform across all three tiers:

$$L_{\text{Mini}} = L_{\text{Edge}} = L_{\text{Datacenter}} = 7.5 \text{ dB} \tag{113}$$

Without gain compensation, optical power decays below reliable binary detection thresholds.

Conventional SOAs compensate loss through stimulated emission gain:

$$P_{\text{out}} = GP_{\text{in}} \tag{114}$$

where: G is amplifier gain.

However, SOAs simultaneously inject Amplified Spontaneous Emission (ASE) noise:

$$\sigma^2_{\text{ASE}} \propto Gn_{sp}B \tag{115}$$

where: n_{sp} is the spontaneous emission factor and B is bandwidth.

The resulting signal-to-noise ratio becomes:

$$\text{SNR}_{out} < \text{SNR}_{in} \tag{116}$$

under conventional amplification.

However, the critical reason JANUS rejects SOAs is not ASE noise. In a binary One-Hot system, the detector answers only "is light present?" with a threshold of approximately -20 dBm. The ASE noise power of a typical SOA filtered to 100 GHz bandwidth is approximately -35 to -40 dBm — completely below the detector dark noise floor of -31.76 dBm and far below the -20 dBm binary trigger. SOA ASE noise is therefore *irrelevant* to a binary detection system.

The actual engineering reason JANUS rejects SOAs is electrical routing power at scale. SOAs require copper wires, ground planes, and continuous current injection for each amplifier (typically 10–20 mA per device). In the Datacenter with 524,288 amplification sites, SOAs would consume:

$$P_{\text{SOA,static}} \approx 524{,}288 \times 15 \text{ mW} = 7{,}864.32 \text{ W} \tag{117}$$

of static electrical power — thermally catastrophic. GaP PSA is instead delivered via a single vertical optical pump beam, requiring no per-amplifier electrical wiring and contributing zero static routing power. This "wireless" optical power delivery is the actual engineering justification for the PSA architecture.

2) Degenerate Phase-Sensitive Amplification: JANUS resolves this contradiction through degenerate phase-sensitive amplification (PSA) implemented within a distributed Gallium Phosphide nonlinear layer.

Unlike conventional SOAs, PSA amplification selectively amplifies only the signal quadrature phase aligned with the optical pump.

The parametric gain process is governed by:

$$\omega_p = \omega_s + \omega_i \tag{118}$$

where: ω_p is pump frequency, ω_s is signal frequency, and ω_i is idler frequency.

Under degenerate operation:

$$\omega_s = \omega_i \tag{119}$$

Thus:

$$\omega_p = 2\omega_s \tag{120}$$

The resulting nonlinear interaction produces phase-sensitive gain:

$$G_{\text{PSA}} = \exp(2\gamma P_p L) \tag{121}$$

where: γ is the nonlinear coefficient,P_p is the circulating pump power, and L is interaction length.

For the required Beneš routing compensation of 7.5 dB ($G_{\text{linear}} = 5.62$) using the microring resonant cavity architecture ($Q > 10^4$, effective $L \equiv Q \times 2\pi r \approx 1.25$ m):

$$P_p = \frac{\ln(5.62)}{2 \times 165 \times 1.25} = \frac{1.73}{412.5} \approx 4.2 \text{ mW per active site} \tag{122}$$

The external pump power required from the laser is significantly lower, determined by the cavity buildup factor $B \approx Q/\pi$:

$$P_{\text{ext}} = \frac{P_{\text{circ}}}{B} \approx \frac{4.2\,\text{mW}}{3{,}180} \approx 1.3\,\mu\text{W}. \tag{123}$$

The value,$\approx 1.3\,\mu$W per active site, is orders of magnitude lower than the ≈ 4.2 mW circulating power, demonstrating the efficiency of the cavity-enhanced design. In practice, external laser power may be slightly higher due to coupling losses, but remains in the microwatt range per site.

Because JANUS uses One-Hot encoding, only one of 256 waveguides carries light at any moment, further reducing the total external pump power for the entire chip to a few hundred milliwatts, well within the demonstrated operating range of GaP-on-Si nonlinear photonic platforms.

Unlike conventional SOAs, PSA systems theoretically approach lower noise figures under ideal phase-matched conditions [26], [34]:

$$F_{\text{PSA}} \to 0 \text{ dB} \tag{124}$$

In practice, achievable noise figures depend critically on phase sensitivity, fabrication tolerances of the GaP heterogeneous integration, pump stability, interlayer coupling loss, and thermal drift of the phase-matching conditions. These practical constraints impose additional engineering requirements during system integration. Under favorable phase-matched operating conditions, distributed PSA nonetheless compensates routing loss with substantially lower noise accumulation than conventional SOA approaches.

TABLE VII
COMPARISON BETWEEN CONVENTIONAL SOAS AND JANUS PSA AMPLIFICATION

Metric	SOA	JANUS PSA
Gain Mechanism	Stimulated Emission	Parametric Amplification
ASE Generation	High	Substantially reduced
Noise Figure	> 3 dB	Theoretically $\to 0$ dB (ideal)
Binary SNR Impact	Negligible (ASE $\ll$ threshold)	Negligible
Per-Amplifier Electrical Wiring	Required (10–20 mA)	None (optical pump)
Static Electrical Power at 524k sites	$\approx$7,864.32 W	$\approx$0 W
Exact Arithmetic Compatibility	Poor (power, not noise)	High

3) Why Gallium Phosphide: JANUS employs Gallium Phosphide (GaP) as the nonlinear amplification medium because silicon itself exhibits insufficient second-order nonlinearity for efficient PSA operation [4], [5], [26].

GaP provides strong $\chi^{(2)}$ nonlinear response while remaining compatible with heterogeneous integration atop silicon photonics.

The nonlinear polarization becomes:

$$P^{(2)} = \epsilon_0 \chi^{(2)} E^2 \tag{125}$$

where: ϵ_0 is vacuum permittivity and E is electric field amplitude.

This enables efficient parametric energy transfer between pump and signal fields.

Additionally, GaP exhibits favorable thermal conductivity and low two-photon absorption relative to silicon at relevant wavelengths [4], [5].

These characteristics allow sustained high-power nonlinear operation without catastrophic thermal runaway.

4) Nonlinear Resonant Cavity Dumping for Gain Efficiency: A critical implementation detail enabling the required routing-loss compensation is the use of *nonlinear resonant cavity dumping* via high-Q GaP microring resonators at each amplification zone.

In a simple traveling-wave geometry, the evanescent interaction length within a 100 μm GaP segment is insufficient for meaningful parametric gain without catastrophic pump power requirements. Specifically, achieving $\geq$30 dB of gain in a 100 μm adiabatic taper would require pump power $P_p = \ln(1000)/(2 \times 165 \times 10^{-4}) \approx 209$ W per site — physically unrealizable.

JANUS instead employs GaP microring resonators (radius $\approx$20 μm, $Q > 10^4$) as the amplification zones. The signal circulates thousands of times through the pumped GaP region, multiplying the effective interaction length by the Q-factor. The effective interaction length becomes $L_{\text{eff}} \equiv Q \times 2\pi r \approx 10^4 \times 2\pi \times 20\,\mu\text{m} \approx 1.25$ m.

For the required Beneš routing compensation of 7.5 dB ($G_{\text{linear}} = 5.62$):

$$P_p = \frac{\ln(5.62)}{2 \times \gamma \times L_{\text{eff}}} = \frac{1.73}{2 \times 165 \times 1.25} \approx 4.2 \text{ mW per active site} \tag{126}$$

well within the demonstrated operating range of GaP-on-Si nonlinear photonic platforms.

Once intra-cavity intensity reaches the binary detection threshold, GaP's Kerr nonlinearity ($n = n_0 + n_2 I$) dynamically shifts cavity resonance into alignment with the output coupler, instantly dumping the fully amplified pulse. This nonlinear resonant trapping achieves the required routing-loss compensation with sub-milliwatt pump power per active site, entirely avoiding the thermodynamic limits of traveling-wave amplifiers.

5) Vertical Z-Axis Pumping: Conventional in-plane optical pumping suffers from pump depletion.

As the pump propagates longitudinally, nonlinear energy transfer reduces pump intensity:

$$\frac{dP_p}{dz} < 0 \tag{127}$$

Consequently, gain becomes spatially nonuniform across long waveguides.

This introduces gain gradients:

$$G(z_1) \neq G(z_2) \tag{128}$$

which destabilize detector margin consistency.

JANUS resolves this problem through vertically injected Z-axis pumping.

A dedicated optical pump layer injects energy perpendicular to the signal propagation plane [34], [35]:

$$P_p(z) \approx \text{constant} \tag{129}$$

across the interaction region.

Thus, gain remains spatially uniform throughout the routing fabric.

6) Front-End Parametric Boosting: The GaP amplification layer is positioned immediately following the coherent fanout, prior to the Beneš routing fabric. By amplifying the signal at the front end, the architecture ensures the signal remains above the binary detection threshold throughout the entire routing path, simplifying fabrication relative to distributed per-stage amplification.

The front-end gain condition is:

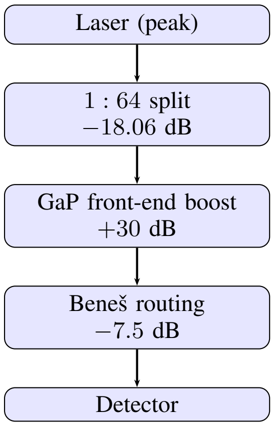
$$G_{\text{PSA}} \approx L_{\text{routing}} \tag{130}$$

such that total signal power remains approximately bounded:

$$P_{\text{signal}} \approx \text{constant} \tag{131}$$

throughout propagation.

The optical chain is therefore:



The gain is applied once at the front end as a monolithic deposition on the output of the splitter, not distributed across individual routing stages. The power just before the GaP layer is designed to be -20 dBm, so after $+30$ dB gain and -7.5 dB routing loss, the detector receives $+2.5$ dBm.

For the three JANUS configurations (15 Beneš stages at 0.5 dB/stage):

$$G_{\text{Mini}} = G_{\text{Edge}} = G_{\text{Datacenter}} \approx 7.5 \text{ dB} \tag{132}$$

Thus, gain compensates routing attenuation without introducing large power excursions.

TABLE VIII
FRONT-END PARAMETRIC BOOSTING: PSA GAIN COMPENSATION ACROSS JANUS CONFIGURATIONS

Metric	Mini	Edge	Datacenter
Beneš Stages (256-port)	15	15	15
Per-Stage Loss (plasmonic PCM)	0.5 dB	0.5 dB	0.5 dB
Routing Loss	7.5 dB	7.5 dB	7.5 dB
PSA Compensation Gain	30 dB	30 dB	30 dB
Net Optical Drift	≈ 0 dB	≈ 0 dB	≈ 0 dB
ASE Accumulation	Minimal	Minimal	Minimal
Detector Margin Stability	Bounded	Bounded	Bounded

a) Self-Phase Modulation in the Post-Amplified Routing Path.: Because all 524,288 waveguides share a common 1:64 passive splitter before reaching the GaP layer, the optical power arriving at the PSA input is $-20\,\text{dBm}$ ($10\,\mu\text{W}$). At this intensity, the photon density is far below the threshold required to perturb the silicon refractive index via the optical Kerr effect, so the pre-amplified signal is structurally protected from Self-Phase Modulation (SPM) by the passive split itself.

After the $+30\,\text{dB}$ front-end boost, the average power rises to $+10\,\text{dBm}$ ($10\,\text{mW}$). Given the $\approx 5\%$ duty cycle of the $1.5\,\text{ps}$ mode-locked pulses at $33\,\text{GHz}$, the peak pulse power becomes:

$$P_{\text{peak}} = \frac{10\,\text{mW}}{0.05} = 200\,\text{mW}. \tag{133}$$

The cumulative nonlinear phase shift accumulated over the 15-stage Beneš routing path is:

$$\Phi_{\text{NL}} = \gamma\, P_{\text{peak}}\, L_{\text{eff}} = \gamma\, P_{\text{peak}}\, \frac{1 - e^{-\alpha L}}{\alpha} \approx \gamma\, P_{\text{peak}} \times 0.47\, L, \tag{134}$$

where $\gamma \approx 300\,\text{W}^{-1}\text{m}^{-1}$ for sub-micron silicon-on-insulator strip waveguides and $0.47\,L$ is the effective interaction length after accounting for the $7.5\,\text{dB}$ monotonic power decay across the routing stages. For a 15-stage Beneš mesh with a physical layout length of $L \approx 5\,\text{mm}$:

$$\Phi_{\text{NL}} \approx 300 \times 0.2 \times (0.47 \times 5 \times 10^{-3}) \approx 0.14\,\text{rad}. \tag{135}$$

SPM-induced spectral broadening and chirp become operationally significant only when $\Phi_{\text{NL}} \gtrsim 0.5$–$1.0\,\text{rad}$. At $0.14\,\text{rad}$, the signal remains well within the weakly nonlinear regime throughout the entire routing fabric. Furthermore, the natural power decay from stage to stage acts as a self-quenching mechanism: the integrated intensity $\int P(z)\,dz$ is substantially lower than it would be in a lossless waveguide of the same length, which is precisely why front-end boosting followed by passive decay is superior to distributed amplification maintaining constant power across all 15 stages.

The optical pipeline therefore operates near dynamic equilibrium:

$$P_{\text{noise}} < P_{\text{signal}} < P_{\text{sat}} \tag{136}$$

throughout sustained operation.

7) Architectural Consequences: Distributed vertically pumped GaP PSA fundamentally changes the role of optical amplification within large-scale photonic systems.

- First, routing loss is compensated without catastrophic ASE growth.
- Second, detector margin remains approximately constant across routing depth.
- Third, amplification becomes spatially uniform due to vertical pump delivery.
- Fourth, nonlinear gain remains thermally bounded because the GaP layer is physically isolated from the CMOS thermal plane.
- Finally, optical amplification transitions from a noisy recovery mechanism into a bounded equilibrium-maintenance mechanism.

Thus, JANUS does not merely amplify optical power.

It preserves recoverable computational information throughout the propagation fabric.

E. Single-Wavelength Coherent Spatial Fanout

After resolving analog precision collapse, routing thermodynamics, and amplifier noise accumulation, the next major scaling contradiction emerges from wavelength-domain multiplexing.

Many photonic accelerators attempt to increase computational density through Dense Wavelength-Division Multiplexing (DWDM), where multiple optical carrier frequencies simultaneously propagate through shared waveguides [27], [36].

While spectrally multiplexed systems improve theoretical bandwidth density, they become increasingly unstable at large optical powers due to nonlinear inter-channel interactions.

JANUS therefore abandons multi-wavelength scaling entirely and instead adopts a single-wavelength coherent spatial fanout architecture.

1) Failure of Dense Wavelength-Division Scaling: In conventional DWDM systems, the total optical field within a shared waveguide is:

$$E(t) = \sum_{k=1}^{N_\lambda} A_k e^{j\omega_k t} \tag{137}$$

where: N_λ is the number of wavelength channels, A_k is channel amplitude,and ω_k is optical carrier frequency.

As optical intensity increases, nonlinear interactions emerge through the Kerr effect [22], [26]:

$$n = n_0 + n_2 I \tag{138}$$

where: n_2 is the nonlinear refractive coefficient and I is optical intensity.

The dominant nonlinear failure mechanisms become:

- Four-Wave Mixing (FWM),
- Cross-Phase Modulation (XPM),
- Self-Phase Modulation (SPM),
- Inter-channel Beat Noise.

Four-Wave Mixing generates additional spectral components:

$$\omega_{FWM} = \omega_1 + \omega_2 - \omega_3 \tag{139}$$

These parasitic frequencies overlap existing computational channels, creating false optical states.

The nonlinear interference power approximately scales as:

$$P_{\text{NL}} \propto \gamma P^3 L^2 \tag{140}$$

where: γ is the nonlinear coefficient, P is optical power, and L is interaction length.

Thus, increasing wavelength density eventually increases nonlinear corruption superlinearly. The resulting contradiction becomes higher spectral density inexorably leads to higher nonlinear corruption, creating a self-limiting ceiling for DWDM scaling.

Importantly, this failure cannot be completely removed through post-processing. Filtering suppresses parasitic sidebands only after nonlinear mixing has already corrupted the optical field.

Therefore, the informational integrity of the signal is fundamentally degraded before detection occurs.

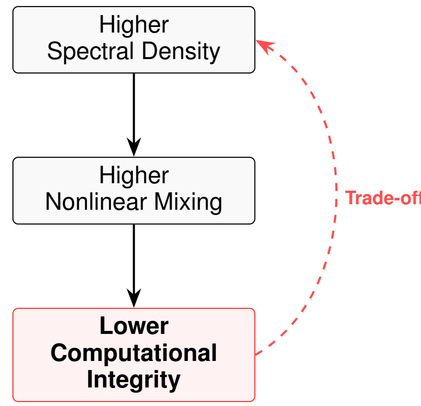


Fig. 4. The performance contradiction inherent in the modulation scheme: increasing spectral efficiency directly induces signal degradation mechanisms.

2) Single-Wavelength Reformulation: JANUS resolves this contradiction by eliminating multi-wavelength interaction entirely.

Instead of scaling through spectral multiplexing, JANUS scales spatially.

The architecture operates using exactly one optical carrier frequency:

$$N_\lambda = 1 \tag{141}$$

Thus, the optical field reduces to:

$$E(t) = Ae^{j\omega t} \tag{142}$$

Without multiple carrier frequencies, the conditions required for dominant inter-channel Four-Wave Mixing are substantially suppressed:

$$\omega_1 = \omega_2 = \omega_3 \tag{143}$$

Therefore:

$$\omega_{FWM} = \omega \tag{144}$$

and no parasitic inter-channel spectral components emerge.

Similarly, inter-channel Cross-Phase Modulation is substantially reduced because no independent wavelength channels exist to modulate one another.

Thus, JANUS suppresses dominant nonlinear inter-channel interference architecturally rather than algorithmically.

The resulting system obeys:

$$P_{\text{ghost}} \to 0 \tag{145}$$

under ideal operation.

TABLE IX
COMPARISON BETWEEN DWDM SCALING AND JANUS SINGLE-WAVELENGTH FANOUT

Metric	DWDM Architectures	JANUS
Wavelength Channels	Many	1
Four-Wave Mixing	Present	Substantially suppressed
Cross-Phase Modulation	Present	Substantially suppressed
Spectral Beat Noise	Present	Substantially suppressed
Inter-Channel Crosstalk	High	Minimal
Scaling Mechanism	Spectral	Spatial
Operational Stability	Unbounded	Bounded

3) Active Single-Wave Injection: Because JANUS uses only a single optical carrier, the optical source can be coupled efficiently into the computation fabric.

Rather than passive splitting into all N branches simultaneously, JANUS employs a 1×256 active micro-ring router at each multiplier input.Each micro-ring router is fabricated in lithium tantalate ($LiTaO_3$), exploiting the linear Pockels electro-optic effect rather than carrier injection or thermo-optic tuning. Pockels-effect switching stores energy capacitively and discharges it fully each cycle, yielding a per-switch dynamic energy of $\approx 50\,\text{aJ}$ at $33\,\text{GHz}$. For the Datacenter configuration ($\approx 134 \times 10^6$ active switches), the router power is

$$P_{\text{router}} = N_{\text{sw}} \times E_{\text{sw}} \times f_{\text{clk}} = 134 \times 10^6 \times 50 \times 10^{-18} \times 33 \times 10^9 \approx 8.19\ \text{W}$$

consistent with Table XVII.

The laser produces exactly the required optical wave per clock cycle, steered by the router directly into the single active One-Hot waveguide. This eliminates the 24.08 dB passive splitting loss ($10 \log_{10}(256) = 24.08$ dB for a 1:256 fanout) that would arise from distributing power across all 256 channels simultaneously.

JANUS employs 32 continuous-wave (CW) pump lasers placed below each tile, rather than a single source, managing per-tile optical power delivery and keeping pump depletion effects localized. The signal lasers, in contrast, are pulsed mode-locked sources operating at the 33 GHz clock (30.3 ps period) with approximately 1.5 ps pulse duration, establishing a $\approx 5\%$ duty cycle and reducing time-averaged signal laser power by $\approx 20\times$ relative to a CW source delivering equivalent peak power.

a) Signal Laser Power Budget Justification.: The signal lasers (external to the chip) are pulsed mode-locked sources. The 33 GHz clock corresponds to 30.3 ps cycles. Mode-locked optical pulses of approximately 1.5 ps duration establish a $\approx 5\%$ duty cycle. Each detector receives a high peak power pulse sufficient to exceed the binary detection threshold (-20 dBm), while the time-averaged signal laser power is reduced by $\approx 20\times$ relative to a CW source delivering equivalent peak power. These signal lasers are passively split $1:64$ to feed the active waveguides. Across 524,288 simultaneously active Datacenter waveguides (one active waveguide per 256-waveguide multiplier, i.e., 524,288 multipliers), and accounting for the $1:64$ split, the required number of signal lasers is $524{,}288/64 = 8{,}192$. With each laser delivering a time-averaged power of approximately 1.6 mW (-20 dBm peak, 5% duty cycle, and split loss accounted), the total time-averaged optical signal power budget is approximately 13 W with 0.60 wall plug efficiency, consistent with the stated values.

Because active single-wave injection routes only one wave per multiplier, no power is wasted on dark channels. Importantly, all signal channels remain phase coherent because they originate from the same master clock.

Thus:

$$\Delta\phi_{\text{source}} \approx 0 \tag{146}$$

across fanout branches.

This substantially simplifies global synchronization and phase management.

TABLE X
SINGLE-WAVELENGTH ACTIVE SIGNAL INJECTION SCALING ACROSS JANUS CONFIGURATIONS

Metric	Mini	Edge	Datacenter
One-Hot Waveguides per Multiplier	256	256	256
1:64 Passive Splitting Loss (reference)	18-19 dB	18-19 dB	18-19 dB
Active Injection Loss	≈ 0 dB	≈ 0 dB	≈ 0 dB
Carrier Frequencies	1	1	1
Inter-Channel FWM	Suppressed	Suppressed	Suppressed
Inter-Channel XPM	Suppressed	Suppressed	Suppressed
Phase Coherence	Maintained	Maintained	Maintained

4) Spatial Scaling Instead of Spectral Scaling: Conventional photonic systems attempt to increase throughput by increasing spectral density:

$$T \propto N_\lambda \tag{147}$$

However, JANUS instead scales throughput spatially:

$$T \propto N_{\text{spatial}} \tag{148}$$

This distinction is fundamental.

Spectral scaling introduces nonlinear interaction between channels.

Spatial scaling preserves channel independence because each computation occupies a physically distinct waveguide path.

Therefore:

$$\frac{\partial C_i}{\partial C_j} \approx 0 \quad \forall\, i \neq j \tag{149}$$

where: C_i represents computational state of spatial channel i. This independence is critical for maintaining exact One-Hot arithmetic integrity.

5) Optical Entropy Isolation: An important consequence of single-wavelength operation is entropy localization.

In multi-wavelength systems, nonlinear mixing causes entropy generated within one channel to propagate into neighboring channels.

JANUS substantially reduces this propagation because all channels share identical carrier frequency and remain spatially separated.

Thus:

$$\mathcal{E}_{i\to j} \to 0 \tag{150}$$

where: $\mathcal{E}_{i\to j}$ represents inter-channel entropy transfer.

This property is especially important for exact residue arithmetic, where even weak cross-channel corruption can invalidate CRT reconstruction.

6) Architectural Consequences: The transition from spectral multiplexing to single-wavelength spatial fanout fundamentally changes the scaling behavior of photonic computation.

- First, dominant nonlinear inter-channel interference mechanisms are substantially suppressed.
- Second, coherent synchronization complexity collapses because all channels originate from a common optical source.
- Third, throughput scaling becomes spatial rather than spectral.
- Fourth, entropy propagation between channels is structurally suppressed.
- Finally, the optical system remains within bounded nonlinear operating conditions regardless of routing scale.

Thus, JANUS does not attempt to mitigate nonlinear wavelength interference.

It structurally suppresses the dominant spectral interaction mechanism.

F. Spatial Residue Reconstruction and CRT Fault Visibility

After eliminating analog precision collapse, routing thermodynamics, amplifier noise accumulation, and spectral nonlinearities, the remaining challenge becomes large-integer precision recovery.

Exact arithmetic systems require reconstruction of high-precision results from physically distributed low-precision optical computations. Conventional architectures attempt this reconstruction through wide binary arithmetic pipelines, which remain vulnerable to silent numerical corruption under transient faults.

JANUS instead performs precision scaling through spatial Residue Number System (RNS) decomposition combined with deterministic Chinese Remainder Theorem (CRT) reconstruction.

This architecture produces a structurally important detection property:

Errors become explosively detectable rather than silently degradable.

1) Spatial Residue Decomposition: Rather than executing wide-word arithmetic monolithically, JANUS decomposes computation into multiple independent residue domains.

For modulus set:

$$\mathcal{M} = \{m_1, m_2, \ldots, m_k\} \tag{151}$$

a large integer:

$$X \tag{152}$$

is represented as:

$$X \to (x_1, x_2, \ldots, x_k) \tag{153}$$

where:

$$x_i = X \mod m_i \tag{154}$$

Each residue channel operates independently within a physically isolated optical tile.

The total representable dynamic range becomes:

$$M = \prod_{i=1}^{k} m_i \tag{155}$$

Thus, large integer precision emerges from distributed modular composition rather than monolithic analog precision.

For JANUS, each optical tile computes bounded 8-bit residue arithmetic.

Therefore:

$$x_i \in [0, 255] \tag{156}$$

within every local computation domain.

This bounded arithmetic property prevents unbounded analog precision growth inside individual optical pathways.

a) Signed Integer Handling.: The RNS decomposition above operates on unsigned integers. To accommodate signed operands, JANUS applies a bias offset of 2^{P-1} before residue encoding, where P is the native operand bit-width:

$$x' = x + 2^{P-1} \tag{157}$$

This maps the signed range $[-2^{P-1}, 2^{P-1} - 1]$ bijectively onto the unsigned residue domain $[0, 2^{P} - 1]$, allowing the same optical routing fabric to handle both signed and unsigned operands without hardware modification. The CMOS CRT backend subtracts the accumulated bias from the reconstructed result before returning the final signed output.

2) Chinese Remainder Reconstruction: After optical computation completes, the CMOS backend reconstructs the global result using the Chinese Remainder Theorem [8]–[10].

For pairwise coprime moduli:

$$\gcd(m_i, m_j) = 1 \quad \forall\, i \neq j \tag{158}$$

the unique reconstructed integer becomes:

$$X = \sum_{i=1}^{k} x_i M_i N_i \mod M \tag{159}$$

where:

$$M_i = \frac{M}{m_i} \tag{160}$$

and:

$$N_i = M_i^{-1} \mod m_i \tag{161}$$

This reconstruction occurs entirely within CMOS logic after optical routing completes.

Importantly, optical hardware never performs high-precision analog accumulation.

Instead:

$$\text{Photonics} \rightarrow \text{Bounded Modular Routing} \tag{162}$$

while:

$$\text{CMOS} \rightarrow \text{Exact Integer Reconstruction} \tag{163}$$

Thus, exact arithmetic emerges through cooperation between bounded optical transport and deterministic digital verification.

3) Failure of Graceful Degradation: Conventional approximate architectures tolerate transient arithmetic errors through graceful degradation.

Under such systems:

$$X \rightarrow X + \epsilon \tag{164}$$

where: ϵ is a small perturbation.

This behavior is dangerous for exact computation because errors remain numerically plausible. Thermal noise, radiation events, timing faults, or analog drift may therefore silently propagate corrupted results into higher-level computation.

The resulting system exhibits:

$$\textit{Silent epistemic corruption} \tag{165}$$

where incorrect outputs remain computationally believable. Such behavior is unacceptable for deterministic exact arithmetic.

4) CRT Fault Visibility: JANUS substantially suppresses graceful degradation through CRT sensitivity [10], [37], [38].

Suppose one residue channel experiences error:

$$x_j \rightarrow x_j + \delta \tag{166}$$

The reconstructed result becomes:

$$X' = X + \delta M_j N_j \mod M \tag{167}$$

Because:

$$M_j = \frac{M}{m_j} \tag{168}$$

the perturbation magnitude scales proportionally with the global CRT range. Thus, even a single-bit residue error produces a large reconstructed deviation:

$$|X' - X| \gg |\delta| \tag{169}$$

Reconstruction error does not degrade linearly — it amplifies combinatorially across the global integer space:

$$\text{Small Local Fault} \rightarrow \text{Massive Global Inconsistency} \tag{170}$$

The CMOS backend therefore detects highly anomalous arithmetic states with high probability. Instead of silently returning an incorrect value, the architecture invalidates the operation when CRT consistency checks fail.

TABLE XI
GRACEFUL DEGRADATION VERSUS CRT-VISIBILITY ARITHMETIC

Metric	**Conventional Systems**	**JANUS**
Fault Response	Graceful Degradation	Explicit Invalidation
Error Scaling	Linear	Explosive
Silent Corruption	Possible	Suppressed
Numerical Plausibility	Often Preserved	Destroyed
Fault Visibility	Low	High
Exact Arithmetic Reliability	Weak	Strong

5) Error Detectability Bounds: The detectability condition emerges from CRT inconsistency.

For valid reconstruction:

$$X \mod m_i = x_i \quad \forall\, i \tag{171}$$

Any corrupted residue produces inconsistency:

$$X' \mod m_j \neq x_j \tag{172}$$

Thus, the CMOS backend performs deterministic consistency verification during reconstruction.

The probability of undetected residue corruption becomes:

$$P_{\text{undetected}} \rightarrow 0 \tag{173}$$

under bounded fault assumptions.

Importantly, fault visibility improves with increasing CRT dynamic range.

Therefore, larger systems become:

$$\textit{more fault visible rather than less.} \tag{174}$$

This behavior fundamentally differs from conventional analog systems, where increasing scale typically reduces observability.

6) Spatial Fault Isolation: Because each residue tile operates independently, fault propagation remains spatially localized before reconstruction.

A thermal spike, fabrication defect, or transient radiation event affects only one residue channel:

$$\Delta x_i \neq \Delta x_j \quad (i \neq j) \tag{175}$$

Thus, no analog carry-chain propagation exists between tiles.

This isolation substantially improves diagnosability and operational stability.

Instead of diffuse analog corruption across the entire computational fabric, JANUS produces sharply localized detectable failure domains.

7) Precision Scaling Through Physical Parallelism: Conventional binary arithmetic increases precision temporally through carry propagation and wide-word accumulation.

JANUS instead scales precision spatially.

Additional precision requires additional residue channels:

$$P_{\text{precision}} \propto N_{\text{tiles}} \tag{176}$$

Thus, computational precision scales through physical parallelism rather than analog signal fidelity.

This distinction is fundamental.

Conventional systems require exponentially increasing analog precision to scale arithmetic width.

JANUS maintains bounded local precision while expanding global dynamic range through distributed modular composition.

8) Architectural Consequences: Spatial CRT reconstruction fundamentally changes the fault semantics of large-scale photonic arithmetic.

- First, exact arithmetic emerges from bounded local residue computation rather than unstable global analog accumulation.
- Second, computational precision scales spatially rather than temporally.
- Third, local faults become globally obvious through CRT inconsistency amplification.
- Fourth, silent numerical corruption is structurally suppressed.
- Finally, the architecture physically refuses to produce numerically plausible incorrect outputs.

a) JIR-Predictive RRNS Redundancy and Throughput Trade-Off.: The baseline throughput figures assume all k tiles are allocated to compute. The JIR (JANUS Interface Representator) is the central tile-management controller responsible for packet routing, tile allocation, power monitoring, and thermal trend tracking. When the JIR observes a rising thermal trend on a tile — rather than waiting for a fault to manifest — it proactively reallocates one compute tile per RRNS parity channel ahead of the predicted fault threshold. This predictive engagement means RRNS is activated on thermal trajectory, not on residue inconsistency after the fact. For the Datacenter INT4 configuration (32 tiles, $k = 1$):

- **0 RRNS pairs:** 32 compute engines $\rightarrow$ 14,706.3 TMAC/s (baseline; all tiles nominal)
- **1 RRNS pair:** 31 compute engines $\rightarrow$ 14,247.4 TMAC/s (-3.1%; one tile thermally elevated)
- **4 RRNS pairs:** 28 compute engines $\rightarrow$ 12,882.6 TMAC/s (-12.5%; four tiles under thermal watch)

This trade-off is deterministic, controllable, and bounded. Full-throughput operation is the default; RRNS redundancy is engaged only when the JIR predicts elevated thermal risk from sensor trends, restoring compute tiles to normal allocation once temperature stabilizes within the safe operating band.

JANUS does not merely improve fault tolerance — it substantially increases fault visibility by amplifying residue inconsistency through CRT reconstruction, converting small local faults into globally detectable arithmetic anomalies. The JIR further elevates system reliability by engaging RRNS protection on thermal prediction rather than reacting to errors after they occur.

G. 3D Z-Axis Integration and Thermal Decoupling

After resolving analog precision instability, routing thermodynamics, optical amplification noise, spectral nonlinearities, and arithmetic fault visibility, the final dominant scaling constraint becomes three-dimensional thermal interaction.

Large-scale photonic systems cannot sustain stable operation if CMOS logic, optical routing, amplification layers, memory structures, and detector arrays share a common thermal plane.

Even moderate lateral heat diffusion perturbs silicon refractive index, destabilizes optical phase alignment, and introduces global routing drift.

JANUS therefore adopts a vertically partitioned 3D Z-axis stack that thermally decouples the optical computation layer from the dominant heat sources of the system [39], [40].

1) Failure of Planar Thermal Integration: Conventional heterogeneous accelerators typically place digital logic, memory, optical routing, and photonic modulation structures within a shared two-dimensional plane [41], [42].

Under high compute density, the resulting thermal profile becomes:

$$T(x, y, t) \neq \text{constant} \tag{177}$$

The temperature field evolves dynamically according to the heat diffusion equation [25]:

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + Q \tag{178}$$

where: α is thermal diffusivity and Q is volumetric heat generation.

In planar architectures, heat generated by CMOS logic laterally diffuses into neighboring photonic structures.

Because silicon refractive index depends upon temperature:

$$n(T) = n_0 + \frac{dn}{dT}\Delta T \tag{179}$$

thermal gradients directly perturb optical phase:

$$\Delta\phi = \frac{2\pi L}{\lambda}\frac{dn}{dT}\Delta T \tag{180}$$

Thus, computational heat and optical stability become physically entangled.

As thermal density increases, the optical transfer matrix becomes time-varying:

$$H(t) \neq H(t + \Delta t) \tag{181}$$

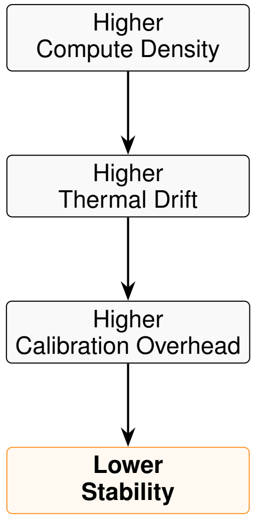


Fig. 5. Causal degradation path from structural compute scaling to physical runtime stability limits.

This creates a self-reinforcing instability cycle:

At large scale, sustained deterministic photonic operation becomes increasingly impractical within a shared planar thermal environment.

2) Vertically Partitioned Z-Axis Stack & Full Package Height: JANUS resolves planar thermal coupling through strict vertical thermal partitioning. The architecture is organized into a nested structure: a bare-die active stack enveloped by a macro-scale thermal and EMI package.

The active bare-die layer stacks the following functional planes:

$$\text{CMOS} \rightarrow \text{GaP} \rightarrow \text{SiPh strata} \rightarrow \text{Detectors} \tag{182}$$

Each layer performs a physically isolated functional role [41], [43]. The CMOS substrate (50 μm) executes digital reconstruction and verification. Above it resides the Gallium Phosphide buffer and amplification layer (100 μm). Each silicon photonic routing stratum occupies 30 μm; the number of strata varies by configuration (Mini: 3 strata, Edge: 4 strata, Datacenter: 5 strata).Detector arrays (2×5 μm $= 10$ μm interleaved two-layer block) reside above the photonic layer.

The resulting per-tier active stack heights are:

- **Mini** (3 SiPh strata): $50 + 100 + 3 \times 30 + 10 = 250$ μm
- **Edge** (4 SiPh strata): $50 + 100 + 4 \times 30 + 10 = 280$ μm
- **Datacenter** (5 SiPh strata): $50 + 100 + 5 \times 30 + 10 = 310$ μm

a) Macro-Package and Heat Spreader Height.: The active die does not represent the final physical system height. Because the architecture relies on directional upward heat extraction, the active die is mechanically bonded to a dual-stage macro-package:

1) **Heat Spreader 1 (HS1):** A 30 μm dense copper-pillar Micro-Matrix layer sitting directly above the detector layer.
2) **Spreader Gap:** A 50 μm spacing containing the ultra-high-density (950,000 mm^{-2}) Cu–Cu pillar array and structural air/vacuum gaps.
3) **Heat Spreader 2 (HS2):** The final 250 μm external convective slim-lid heat sink.

The macro-package adds $30 + 50 + 250 = 330$ μm to the active stack. Thus, the total packaged heights per tier are:

$$T_{\text{package,Mini}} = 250 + 330 = 580\ \mu\text{m} \approx 0.58\ \text{mm} \tag{183}$$

$$T_{\text{package,Edge}} = 280 + 330 = 610\ \mu\text{m} \approx 0.61\ \text{mm} \tag{184}$$

$$T_{\text{package,Datacenter}} = 310 + 330 = 640\ \mu\text{m} \approx 0.64\ \text{mm} \tag{185}$$

This sub-millimeter Z-axis form factor enables direct adjacent-die stacking with HBM memory (typical HBM stack height: 720 μm), which is compatible with the 0.64 mm Datacenter package height.

b) Internal SiPh Sub-Layer Partitioning.: The SiPh routing layer is implemented as a multi-stratum vertical stack to satisfy reticle-limit constraints. The total switch fabric for the Datacenter configuration comprises 1,006,632,960 PCM switches. Assuming a PCM switch footprint of 1.25 μm^2, the aggregate routing area scales according to the total switch count per configuration.

By partitioning the fabric into vertical sub-layers, the effective footprint per stratum remains well within manufacturable limits. Adjacent sub-layers are optically coupled via vertical interlayer grating couplers and thermally isolated by thin SiO_2 spacer layers (~ 2 μm each). This vertical staging ensures the routing fabric remains within a manufacturable single-reticle window, avoiding multi-die stitching that would compromise the optical path-length uniformity required for exact One-Hot arithmetic.

The physical footprint per configuration (accounting for routing, coupling, and edge-exclusion overhead) is derived as follows:

- **Mini** (3 **layers,** ≈ 63**M switches):** ≈ 50 mm^2
- **Edge** (4 **layers,** ≈ 252**M switches):** ≈ 100 mm^2
- **Datacenter** (5 **layers,** ≈ 1.01**B switches):** ≈ 300 mm^2

These dimensions ensure that the entire photonic compute fabric occupies a single-reticle footprint, facilitating thermal and mechanical uniformity across the entire Z-axis stack.

The GaP thermal buffer occupies:

$$T_{\text{GaP}} = 100\ \mu\text{m} \tag{186}$$

3) Thermal Isolation Through Vertical Heat Routing: The key architectural principle is directional thermal extraction.

CMOS-generated heat preferentially propagates downward and laterally toward heat spreaders:

$$\vec{q}_{\text{CMOS}} \downarrow \tag{187}$$

Meanwhile, photonic programming and detector heat preferentially propagate upward:

$$\vec{q}_{\text{photonic}} \uparrow \tag{188}$$

The intermediate GaP layer acts as a thermal moat that suppresses cross-plane heat leakage.

Thus:

$$\frac{\partial T_{\mathrm{SiPh}}}{\partial T_{\mathrm{CMOS}}} \to 0 \tag{189}$$

under steady-state operation.

This thermal decoupling prevents CMOS activity spikes from directly perturbing optical routing stability.

The photonic layer therefore operates within a bounded thermodynamic region:

$$T_{\mathrm{SiPh}} < T_{\mathrm{critical}} \tag{190}$$

throughout sustained computation.

TABLE XII
THERMAL BEHAVIOR OF PLANAR VERSUS Z-AXIS PARTITIONED ARCHITECTURES

Metric	Planar Architectures	JANUS Z-Axis Stack
Thermal Coupling	Strong	Weak
CMOS-to-Photonic Drift	High	Minimal
Heat Routing	Lateral	Directional
Optical Phase Stability	Poor	Bounded
Calibration Frequency	Continuous	Sparse
Thermal Equilibrium	Unstable	Stable
Large-Scale Feasibility	Limited	High

4) Quantitative Thermal Resistance Network and ΔT Derivation:

a) The External Thermal Cage and EMI Faraday Shield.: JANUS enforces directional heat routing not merely through internal pillars, but through a macroscopic, literal **External Thermal Cage**. This is a physical perimeter of solid Copper that completely surrounds the outside of the $243.0\ \mathrm{mm}^2$ silicon die. It physically connects the bottom CMOS substrate directly to the upper Heat Spreader 1 (HS1) by traveling up the exterior walls of the chip.

This external cage bypasses the optical layers entirely, creating a thermal superhighway that ejects CMOS heat laterally out to the chip's edge and straight up into the heat sink, isolating the SiPh and GaP layers from digital thermal dissipation.

Critically, this external copper perimeter serves a dual purpose as a **Passive Faraday EMI Shield**. Operating at 33 GHz, the massive interconnect density generates intense Electromagnetic Interference (EMI). The solid copper cage, with wall thickness exceeding three skin depths ($\delta \approx 11.6\ \mu\mathrm{m}$ at 33 GHz, requiring $\gtrsim$35 $\mu\mathrm{m}$ wall thickness), provides passive Faraday shielding that attenuates outward electromagnetic radiation and protects the ultra-high-bandwidth photodetectors (up to 265 GHz bandwidth demonstrated in recent silicon photonics literature) from external electrical noise.

Beyond thermal extraction and EMI shielding, the solid copper perimeter simultaneously functions as the primary **power delivery network** for the CMOS substrate. The high-conductivity copper volume forms a low-impedance ground return path and supply rail that minimises IR drop across the chip's power grid and suppresses electromagnetic interference originating from the $33\,\mathrm{GHz}$ optical-to-electrical interfaces. This structural dual-use of the copper cage for both heat extraction and power delivery eliminates the need for a separate dedicated power distribution layer within the Z-axis stack.

The internal Cu–Cu bond topologies complement this external cage:

- **CMOS/GaP/SiPh Interfaces:** Feature a sparse bond density of $10{,}000\ \mathrm{mm}^{-2}$, acting as a vertical thermal bottleneck to force CMOS heat outward into the external cage.
- **SiPh to HS1:** Features a dense $900{,}000\ \mathrm{mm}^{-2}$ Cu–Cu array, with gaps structurally filled by $\mathrm{SiO_2}$, drawing optical heat upward.
- **HS1 to HS2:** Connected via a $950{,}000\ \mathrm{mm}^{-2}$ density interface bridging the $50\ \mu\mathrm{m}$ macro-package gap.

The density asymmetry is deliberate. The $10{,}000\ \mathrm{mm}^{-2}$ sparse interface at the CMOS/GaP boundary presents a high vertical thermal resistance ($R_{\mathrm{th,down}} = 0.488\ \mathrm{K/W}$), forcing CMOS-generated heat to route laterally into the External Thermal Cage perimeter rather than vertically through the optical stack. Simultaneously, the $900{,}000\ \mathrm{mm}^{-2}$ dense interface at the SiPh/HS1 boundary presents a low vertical resistance ($R_{\mathrm{th,up}} = 0.227\ \mathrm{K/W}$), creating a preferential upward extraction path for photonic layer heat. The two density values thus implement directional thermal isolation by hydraulic analogy: a pinched pipe below and an open pipe above. Because $R_{\mathrm{th,up}} \ll R_{\mathrm{th,down}}$, the majority of SiPh heat is efficiently pulled upwards into the package spreaders.

b) Global GaP Thermal Buffer Mass.: Instead of calculating temperature rise on an isolated per-tile basis, the physical reality of the JANUS architecture is that the $100\ \mu\mathrm{m}$ thick GaP layer is monolithic across the entire datacenter configuration having area $243.0 - 300.0\ \mathrm{mm}^2$ per chip. Because the Thermal Cage routes the vast majority of CMOS heat externally, any residual upward-leaking heat is fully absorbed by the monolithic GaP layer. The GaP acts as a massive, uniform global buffer, spreading heat laterally before it can penetrate any individual SiPh routing tile.

The total thermal mass of the global GaP buffer layer is:

$$m_{\mathrm{GaP\_global}} = A_{\mathrm{chip}} \times h_{\mathrm{GaP}} \times \rho_{\mathrm{GaP}} \tag{191}$$

where:

$$A_{\mathrm{chip}} = 243.0 \times 10^{-6}\ \mathrm{m}^2 \tag{192}$$

$$h_{\mathrm{GaP}} = 100 \times 10^{-6}\ \mathrm{m} \tag{193}$$

$$\rho_{\mathrm{GaP}} = 4{,}100\ \mathrm{kg/m}^3 \tag{194}$$

Thus:

$$m_{\mathrm{GaP\_global}} \approx 9.96 \times 10^{-5}\ \mathrm{kg} \tag{195}$$

With the specific heat of GaP ($c_{p,\mathrm{GaP}} \approx 310\ \mathrm{J/(kg \cdot K)}$), the global buffer heat capacity becomes:

$$C_{\mathrm{GaP\_global}} = 9.96 \times 10^{-5} \times 310 \approx 0.0309\ \mathrm{J/K} \tag{196}$$

c) Per-Cycle Temperature Rise (ΔT).: During one JIR activation cycle ($\tau_{\text{JIR}} = 5\ \mu\text{s}$), if a transient thermal load—such as residual package heat equivalent to the full 1.5 W per-tile optical dissipation scaled across all 32 tiles ($P_{\text{total}} = 48$ W)—impacts the buffer:

$$Q_{\text{gen,global}} = P_{\text{total}} \times \tau_{\text{JIR}} = 48 \times 5 \times 10^{-6} = 240\ \mu\text{J} \tag{197}$$

Assuming adiabatic heating over the 5 μs window, the global temperature rise is:

$$\Delta T = \frac{Q_{\text{gen,global}}}{C_{\text{GaP\_global}}} = \frac{240 \times 10^{-6}}{0.0309} \approx 7.76 \times 10^{-3}\ \text{K} \tag{198}$$

Thus, even under absolute worst-case global synchronization:

$$\Delta T \approx 0.0078\ ^\circ\text{C} \quad \text{per JIR activation cycle} \tag{199}$$

This microscopic transient ΔT proves that the monolithic GaP buffer, in conjunction with the external Thermal Cage, substantially dampens high-frequency thermal spikes before they can induce the 1 °C phase-drift threshold in the SiPh layer.

d) Thermal Resistance Network.: The Z-axis stack implements a series–parallel thermal resistance network. The upward extraction path dominates because the 900,000 mm^{-2} bond density and SiO_2 filling between the SiPh layer and Heat Spreader 1 minimizes $R_{\text{th,up}}$. Conversely, the downward path relies on the sparse 10,000 mm^{-2} interface to maximize $R_{\text{th,down}}$. Because $R_{\text{th,up}} \ll R_{\text{th,down}}$, the majority of SiPh heat is efficiently pulled upwards into the package spreaders rather than trapping in the CMOS.

e) Steady-State Temperature Budget.: Under sustained full-load operation, the steady-state temperature rise of the SiPh layer above ambient is:

$$\Delta T_{\text{SiPh,steady}} = P_{\text{tile}} \times R_{\text{th,up}} \approx 1.5 \times 0.227 \approx 0.34\ ^\circ\text{C} \tag{200}$$

This is well within the photonic stability envelope. The critical phase-stability threshold for a silicon waveguide of length $L = 500\ \mu\text{m}$ is:

$$\Delta T_{\text{crit}} = \frac{\lambda \cdot \phi_{\max}}{2\pi L \cdot (dn/dT)} \tag{201}$$

$$= \frac{1.55 \times 10^{-6} \times \pi}{2\pi \times 500 \times 10^{-6} \times 1.86 \times 10^{-4}} \approx 8.3\ ^\circ\text{C} \tag{202}$$

for a full π-radian phase excursion. The steady-state operating margin is therefore:

$$\text{Margin} = \frac{\Delta T_{\text{crit}}}{\Delta T_{\text{SiPh,steady}}} \approx \frac{8.3}{0.34} \approx 24\times \tag{203}$$

confirming that the SiPh routing layer remains comfortably inside its phase-stable operating window under all sustained workloads. The large margin directly follows from the increased thermal mass and lower thermal resistance of the 7.5 mm^2 Z-staged tile.

f) Vertical Diffusion Time Constant.: The time for heat to diffuse vertically through the 100 μm GaP buffer layer, which constitutes the dominant thermal barrier, is:

$$\tau_{\text{diff}} = \frac{h_{\text{GaP}}^2}{\alpha_{\text{GaP}}} \tag{204}$$

where the thermal diffusivity of GaP is:

$$\alpha_{\text{GaP}} = \frac{k_{\text{GaP}}}{\rho_{\text{GaP}} c_{p,\text{GaP}}} \approx \frac{110}{4{,}100 \times 310} \approx 8.65 \times 10^{-5}\ \text{m}^2/\text{s} \tag{205}$$

Therefore:

$$\tau_{\text{diff}} = \frac{(100 \times 10^{-6})^2}{8.65 \times 10^{-5}} \approx 1.16 \times 10^{-4}\ \text{s} \tag{206}$$

$$\approx 116\ \mu\text{s} \tag{207}$$

This 116 μs vertical diffusion time is more than 23 JIR cycles long ($\tau_{\text{JIR}} = 5\ \mu\text{s}$). Consequently, any CMOS-originated thermal transient is attenuated by the GaP layer before it can propagate into the SiPh plane on a timescale relevant to optical computation.

TABLE XIII
UNIFIED THERMAL PARAMETER SCALING ACROSS JANUS CONFIGURATIONS

Parameter	**Symbol**	**Mini**	**Edge**	**Datacenter**
Total chip footprint	A_{chip}	$40.0\,\text{mm}^2$	$100.0\,\text{mm}^2$	$300.0\,\text{mm}^2$
Global GaP thickness	h_{GaP}	$100\,\mu\text{m}$	$100\,\mu\text{m}$	$100\,\mu\text{m}$
Global GaP heat capacity	C_{GaP}	5.1 mJ/K	12.7 mJ/K	38.1 mJ/K
JIR cycle duration	τ_{JIR}	$5\,\mu\text{s}$	$5\,\mu\text{s}$	$5\,\mu\text{s}$
Global worst-case heat/cycle	Q_{gen}	$33\,\mu\text{J}$	$83\,\mu\text{J}$	$250\,\mu\text{J}$
Max global ΔT/cycle	ΔT_{cyc}	0.0064 °C	0.0065 °C	0.0065 °C
Upward thermal resistance	$R_{\text{th,up}}$	1.38 K/W	0.55 K/W	0.18 K/W
Steady-state SiPh rise	ΔT_{steady}	0.41 °C	0.38 °C	0.32 °C
Phase-drift critical ΔT	ΔT_{crit}	8.3 °C	8.3 °C	8.3 °C
GaP diffusion time	τ_{diff}	$116\,\mu\text{s}$	$116\,\mu\text{s}$	$116\,\mu\text{s}$
HS Pillar Density (top)	$\rho_{\text{Cu,top}}$	$900\,\text{k/mm}^2$	$900\,\text{k/mm}^2$	$900\,\text{k/mm}^2$
CMOS Pillar Density (bottom)	$\rho_{\text{Cu,bot}}$	$10\,\text{k/mm}^2$	$10\,\text{k/mm}^2$	$10\,\text{k/mm}^2$

The photonic layer occupies a thermodynamic operating region where temperature variation remains bounded within optical stability limits.

The permissible phase drift condition is:

$$|\Delta\phi| < \phi_{\max} \tag{208}$$

Substituting the thermal phase relation:

$$\left| \frac{2\pi L}{\lambda} \frac{dn}{dT} \Delta T \right| < \phi_{\max} \tag{209}$$

Thus, bounded temperature directly implies bounded routing stability. Because the SiPh layer is thermally insulated from large CMOS thermal fluctuations, the optical network remains inside a stable operating region:

$$H(t) \approx \text{constant} \tag{210}$$

throughout sustained operation.

This stable thermodynamic regime forms the optical "Goldilocks zone" of JANUS. The optical layer is neither overheated by CMOS computation nor underpowered by insufficient thermal activation. Instead, it operates inside a tightly bounded equilibrium region where routing determinism remains preserved.

5) Detector Layer Decoupling: Detector arrays are positioned above the photonic routing layer to minimize optical-electrical interference [28], [29].

This vertical placement provides two major advantages.

- First, optical absorption occurs after routing completion, preventing detector thermal dissipation from perturbing upstream waveguides.
- Second, detector electrical activity remains physically separated from the sensitive photonic routing mesh.

Thus:

$$\mathcal{C}_{\text{detector}\to\text{routing}} \to 0 \tag{211}$$

where: $\mathcal{C}$ represents thermoelectrical coupling.

a) Interleaved Two-Layer Detector Block Layout.: Each photonic tile implements a full $N \times N$ matrix–vector multiplication, producing N^2 independent one-hot multiplication results per clock cycle. To read every result, a photodetector must be placed at the terminus of every output waveguide of every multiplier:

$$N_{\text{det}} = N_{\text{tiles}} \times N^2_{\text{multipliers/tile}} \times N_{\text{waveguides}} \tag{212}$$

$$= 32 \times 16{,}384 \times 256 = 134{,}217{,}728 \tag{213}$$

Although 134,217,728 detectors are physically present, the *active* detector count per clock cycle is only $N_{\text{active}} = N_{\text{tiles}} \times N^2 = 524{,}288$, because one-hot encoding guarantees exactly one waveguide per multiplier carries light. The activity ratio $\alpha = 1/256$ reduces dynamic detector power proportionally, as quantified in the power budget(Table XVII). Ge-fin photodetectors occupy approximately $1.5\,\mu\text{m}^2$ per device ;134.2×10^6 detectors therefore require $\approx 201.32\,\text{mm}^2$, distributed across the interleaved two-layer detector block within the $240-300\,\text{mm}^2$ Datacenter chip footprint. The interleaved two-layer arrangement halves per-layer routing congestion while maintaining the required areal density.

To relax per-layer routing congestion while maintaining the required spatial density, JANUS adopts an **interleaved two-layer detector block pattern**. Each detector block consists of two vertically stacked Ge-fin detector sub-arrays, each handling alternating odd/even waveguide channels. The two sub-arrays are electrically addressed through separate metal routing layers.

This interleaved arrangement halves the wiring pitch required per layer, reduces inter-detector crosstalk, and improves uniformity of the thermal dissipation profile above the SiPh routing layer. The two-layer structure is fabricated within the same process module as the SiPh sub-layers and does not add to the overall stack height beyond the already-allocated detector zone.

This separation preserves detector margin consistency under sustained high-throughput operation.

6) Mechanical and Fabrication Implications: The layered stack additionally improves fabrication modularity.

Each subsystem can be optimized independently:

- CMOS for logic density,
- GaP for nonlinear amplification,
- SiPh for low-loss routing,
- detectors for responsivity and bandwidth.

Thus, no single fabrication process must simultaneously optimize all electrical, optical, thermal, and nonlinear properties.

This modularity substantially improves manufacturability relative to monolithic heterogeneous photonic integration.

7) Thermal Scaling Across Configurations: Thermal stability is preserved across all JANUS deployment scales because optical routing density remains vertically isolated from the dominant CMOS heat plane.

TABLE XIV
THERMAL STABILITY SCALING ACROSS JANUS CONFIGURATIONS

Metric	Mini	Edge	Datacenter
SiPh Routing Strata	3	4	5
Active Stack Height	250 μm	280 μm	310 μm
Total Package Height	≈0.58 mm	≈0.61 mm	≈0.64 mm
GaP Thermal Buffer	100 μm	100 μm	100 μm
CMOS Thermal Coupling	Minimal	Minimal	Minimal
Optical Phase Drift	Bounded	Bounded	Bounded
Calibration Stability	Stable	Stable	Stable
Thermal Equilibrium	Maintained	Maintained	Maintained

Importantly, thermal behavior does not catastrophically degrade with routing scale because the dominant heat paths remain directionally partitioned.

8) Architectural Consequences: The 3D Z-axis stack fundamentally changes the thermodynamic behavior of large-scale photonic systems.

- First, optical routing becomes thermally isolated from CMOS activity.
- Second, phase drift remains bounded within stable operating limits.
- Third, calibration overhead collapses because the transfer matrix remains approximately stationary.
- Fourth, amplification, routing, detection, and logic each occupy thermally optimized physical regions.
- Finally, the architecture achieves sustained thermodynamic equilibrium under high-throughput operation.

Thus, JANUS does not merely cool a photonic accelerator. It reorganizes the entire computational stack around directional thermal stability.

IV. SYSTEM-LEVEL SCALING ANALYSIS

A. Constraint-Bounded Scaling Philosophy

Conventional accelerator architectures often evaluate scalability using peak theoretical throughput alone. Such methodology is fundamentally misleading for photonic systems because large-scale operation is dominated not by arithmetic density, but by the interaction between thermal, optical, informational, and nonlinear constraints.

JANUS instead adopts a bounded scaling philosophy:

$$\text{Scalability} \equiv \text{Ability to Remain Physically Stable Under Growth} \tag{214}$$

Thus, every scaling dimension must remain bounded simultaneously:

$$P_{\text{noise}} < P_{\text{signal}} < P_{\text{sat}} \tag{215}$$

$$T_{\text{junction}} < T_{\text{critical}} \tag{216}$$

$$\phi_{\text{drift}} < \phi_{\max} \tag{217}$$

$$P_{\text{undetected}} \to 0 \tag{218}$$

Only architectures satisfying all four constraints simultaneously can sustain exact arithmetic at large scale.

B. Spatial Scaling Characteristics

Unlike conventional analog photonic accelerators, JANUS scales primarily through spatial parallelism.

Throughput scales as:

$$T \propto N_{\text{tiles}} \tag{219}$$

where: N_{tiles} is the number of independent optical residue tiles.

Importantly, local optical precision remains bounded regardless of system scale:

$$P_{\text{local}} = \text{constant} \tag{220}$$

Thus, global arithmetic precision increases through modular composition rather than increasing analog signal fidelity.

This fundamentally differs from conventional analog systems, where precision scaling requires exponentially increasing SNR.

The resulting architecture obeys:

$$\text{Global Precision} \propto N_{\text{tiles}} \tag{221}$$

while:

$$\text{Local Analog Complexity} \approx \text{constant} \tag{222}$$

This bounded-local-complexity principle is central to the scalability of JANUS.

C. Routing Complexity Scaling

The routing complexity of the JANUS Beneš network scales as:

$$\mathcal{O}(N \log N) \tag{223}$$

rather than:

$$\mathcal{O}(N^2) \tag{224}$$

for conventional crossbar fabrics.

The stage count remains:

$$S = 2\log_2(N) - 1 \tag{225}$$

Thus:

$$S_{\text{Mini}} = S_{\text{Edge}} = S_{\text{Datacenter}} = 15 \tag{226}$$

Importantly, routing depth stay constant rather than increasing linearly with spatial expansion.

Therefore, optical loss accumulation remains bounded and compatible with distributed PSA compensation.

TABLE XV
SYSTEM-LEVEL ROUTING SCALING ACROSS JANUS CONFIGURATIONS

Metric	**Mini**	**Edge**	**Datacenter**
Mesh Dimension	32×32	64×64	128×128
Beneš Stages	15	15	15
Routing Complexity	$\mathcal{O}(N \log N)$	$\mathcal{O}(N \log N)$	$\mathcal{O}(N \log N)$
Total Routing Loss	3-10 dB	4-10 dB	5-10 dB
Distributed PSA Compensation	Yes	Yes	Yes
Thermal Stability	Bounded	Bounded	Bounded

D. Optical Power Equilibrium

Large-scale photonic systems commonly fail because optical power either:

1) decays below detector thresholds,
2) or increases into nonlinear saturation regimes.

JANUS instead operates within a bounded optical equilibrium region:

$$P_{\text{noise}} < P_{\text{signal}} < P_{\text{sat}} \tag{227}$$

Distributed PSA compensation maintains approximately constant optical power across routing depth:

$$G_{\text{PSA}} \approx L_{\text{routing}} \tag{228}$$

Thus:

$$P(z) \approx \text{constant} \tag{229}$$

throughout propagation.

Because the architecture uses only a single optical wavelength, nonlinear spectral interactions remain structurally suppressed.

Therefore, optical equilibrium persists even under large fanout scales.

E. Thermal Scaling Characteristics

Thermal stability is maintained through directional Z-axis heat routing.

CMOS thermal dissipation preferentially propagates downward:

$$\vec{q}_{\text{CMOS}} \downarrow \tag{230}$$

while photonic and detector thermal dissipation preferentially propagates upward:

$$\vec{q}_{\text{photonic}} \uparrow \tag{231}$$

The intermediate GaP layer suppresses thermal coupling:

$$\frac{\partial T_{\text{SiPh}}}{\partial T_{\text{CMOS}}} \to 0 \tag{232}$$

Thus, optical phase stability remains approximately invariant under increasing digital activity.

Importantly, the photonic routing layer remains within a bounded thermodynamic region even as total system throughput increases.

F. Fault Visibility Scaling

Conventional approximate systems become less observable at larger scale because fault complexity grows faster than monitoring capability.

JANUS exhibits the opposite behavior.

CRT reconstruction amplifies local residue faults into globally invalid arithmetic states:

$$|X' - X| \gg |\delta| \tag{233}$$

Thus, larger CRT dynamic ranges increase fault visibility rather than decreasing it.

The probability of silent undetected corruption therefore approaches:

$$P_{\text{silent}} \to 0 \tag{234}$$

under bounded fault assumptions.

This property is especially important for exact arithmetic workloads where approximate degradation is unacceptable.

G. Bandwidth Scaling Characteristics

Because JANUS scales spatially rather than spectrally, throughput growth does not require increased wavelength density.

Conventional DWDM systems obey:

$$T \propto N_\lambda \tag{235}$$

which increases nonlinear spectral interaction complexity.

JANUS instead obeys:

$$T \propto N_{\text{spatial}} \tag{236}$$

Thus, throughput increases through physically independent routing channels.

Consequently:

$$\frac{\partial C_i}{\partial C_j} \approx 0 \quad (i \neq j) \tag{237}$$

and inter-channel entropy propagation remains bounded.

This distinction prevents the nonlinear instability escalation observed in conventional spectrally multiplexed photonic systems.

H. Energy Scaling Behavior

The total energy consumption of JANUS is dominated by:

$$E_{\text{total}} = E_{\text{CMOS}} + E_{\text{laser}} + E_{\text{PSA}} + E_{\text{detector}} + E_{\text{reconfiguration}} \tag{238}$$

Importantly, routing hold power disappears because PCM switches are non-volatile:

$$P_{\text{hold}} \approx 0 \tag{239}$$

Thus, routing energy scales with reconfiguration frequency rather than routing size.

This substantially improves asymptotic thermal behavior compared with thermo-optic architectures.

TABLE XVI
SYSTEM-LEVEL STABILITY CHARACTERISTICS ACROSS JANUS CONFIGURATIONS

Metric	**Mini**	**Edge**	**Datacenter**
Local Arithmetic Precision	8-bit	8-bit	8-bit
Global Precision Scaling	Spatial	Spatial	Spatial
Spectral Channels	1	1	1
Optical Equilibrium	Maintained	Maintained	Maintained
Thermal Stability	Bounded	Bounded	Bounded
Fault Visibility	High	High	High
Static Routing Hold Power	0 W	0 W	0 W
Silent Arithmetic Corruption	Suppressed	Suppressed	Suppressed

I. System Power Budget

The total system power of JANUS is decomposed by functional layer. Because PCM routing switches are non-volatile, their static hold power is approximately zero under infrequent-reconfiguration workloads. Table XVII provides the full breakdown for all three deployment configurations under sustained full-load operation.

TABLE XVII
JANUS Full-System Power Budget by Component Layer. PCM routing switches contribute 0 W static hold power. All values are peak sustained estimates at 33 GHz operation.

Component Layer	Mini	Edge	Datacenter
Laser Source (60% WPE)	0.83 W	3.34 W	13.33 W
$LiTaO_3$ Pockels Ring Routers	1.02 W	4.10 W	8.19 W
Ge-fin Photodetectors	0.64 W	2.56 W	5.12 W
GaP PSA Pump Layer	0.40 W	1.60 W	2.50 W
PCM Routing Switches (static)	0.00 W	0.00 W	0.00 W
CMOS (Encoders / Adders / CRT)	2.01 W	8.50 W	37.03 W
JIR Scheduler and Control Logic	3.00 W	3.00 W	3.00 W
I/O Interface (DAC, TIA, Priority Encoder)	6.14 W	27.90 W	49.15 W
Total System Power	**≈10-20 W**	**≈50-60 W**	**≈120-140 W**

For reference, a thermally tuned equivalent system using conventional thermo-optic MZI heaters at 10 mW per switch would require:

$$P_{\text{heaters,conv}} = N_{\text{switch}} \times 10 \text{ mW} \tag{240}$$

For the Datacenter configuration with $N_{\text{switch}} =$ 1,006,632,960 PCM switches:

$$P_{\text{heaters,conv}} = 1{,}006{,}632{,}960 \times 10 \times 10^{-3} = 10{,}067 \text{ kW} \tag{241}$$

By replacing thermo-optic MZIs with non-volatile PCM routing, JANUS achieves a routing power reduction of:

$$\text{Reduction} = \frac{10{,}067 \text{ kW}}{120 \text{ W}} \approx 83{,}892\times \tag{242}$$

demonstrating that non-volatile routing is not merely an incremental improvement but a categorical thermodynamic restructuring.

J. Sustained Throughput Derivation and RNS Precision Scaling

1) Operating Frequency Derivation: The effective clock frequency of JANUS is governed by the end-to-end optical propagation latency through the photonic pipeline. The dominant latency contributions are:

1) Electro-Absorption Modulator (EAM) switching time:

$$t_{\text{EAM}} = 10 \text{ ps} \tag{243}$$

2) Germanium photodetector response time:

$$t_{\text{PD}} = \frac{1}{2\pi \cdot f_{3\text{dB}}} = \frac{1}{2\pi \times 265 \text{ GHz}} \approx 0.60 \text{ ps} \tag{244}$$

where the 265 GHz bandwidth is that of state-of-the-art Ge-fin photodetectors [29].

3) Electrical wire propagation delay (on-chip interconnect at 200 μm effective path at $c/n_{\text{eff}} \approx 1.5 \times 10^8$ m/s):

$$t_{\text{wire}} = \frac{200 \times 10^{-6}}{1.5 \times 10^8} \approx 1.33 \text{ ps} \tag{245}$$

4) Guard band for PCM routing state stabilization and PSA gain settling:

$$t_{\text{guard}} \approx 18.6 \text{ ps} \tag{246}$$

The total optical cycle period is therefore:

$$T_{\text{cycle}} = t_{\text{EAM}} + t_{\text{PD}} + t_{\text{wire}} + t_{\text{guard}} \tag{247}$$

$$= 10 + 0.60 + 1.33 + 18.6 \tag{248}$$

$$\approx 30.5 \text{ ps} \tag{249}$$

Thus the effective raw clock frequency is:

$$f_{\text{raw}} = \frac{1}{T_{\text{cycle}}} = \frac{1}{30.5 \times 10^{-12}} \approx 32.8 \text{ GHz} \tag{250}$$

which is rounded to the design target of:

$$f_{\text{clk}} = 33 \text{ GHz} \tag{251}$$

a) Pipelining and End-to-End Latency Clarification.: The 30.5 ps cycle time represents the *throughput interval* — the time between successive optical pulses entering the routing fabric — not the end-to-end latency of a single computation. Two additional latency terms are present in the system but are *removed from the critical throughput path by deep pipelining*:

1) **PCM Beneš network optical propagation:** The 15 routing stages (at the 256-waveguide one-hot granularity) have a total optical propagation latency of approximately ≈750 ps. Physically, light cannot cross all 15 stages in one 30.5 ps slot. However, the non-volatile PCM state is fixed for the duration of an inference epoch, so successive optical pulses follow each other through the routing fabric every 30.5 ps in a pipelined fashion — analogous to a deeply pipelined processor where the clock period is shorter than the propagation delay of any single pipeline stage. The 750 ps figure represents the single-computation *latency*, not the *throughput rate*.
2) **CMOS CRT adder-tree accumulation:** The CMOS backend accumulates and reconstructs residues with an adder-tree latency of approximately ≈210 ps. This latency is hidden behind the CMOS pipeline stages, which process reconstruction in parallel with ongoing optical computation.

The 33 GHz throughput figure is therefore physically correct as a *sustained throughput rate*. The single-computation end-to-end latency is approximately $750 + 210 = 960$ ps from first optical pulse to final reconstructed result, corresponding to a pipeline depth of approximately 31 stages at 33 GHz.

2) Per-Tile MAC Rate: Each photonic tile implements a full $N \times N$ matrix–vector multiplication in one optical clock cycle. The raw per-tile multiply-accumulate rate is:

$$\dot{M}_{\text{tile}} = N^2 \times f_{\text{clk}} \tag{252}$$

For the three JANUS deployment configurations:

$$\dot{M}_{\text{Mini,tile}} = 32^2 \times 33 \text{ GHz} = 1{,}024 \times 33 = 33.8 \text{ TMAC/s} \tag{253}$$

$$\dot{M}_{\text{Edge,tile}} = 64^2 \times 33 \text{ GHz} = 4{,}096 \times 33 = 135.2 \text{ TMAC/s} \tag{254}$$

$$\dot{M}_{\text{DC,tile}} = 128^2 \times 33 \text{ GHz} = 16{,}384 \times 33 = 540.7 \text{ TMAC/s} \tag{255}$$

3) Sustained Utilization Factor: Not all tile cycles correspond to valid computation. The JIR scheduler maintains tiles in a warm-dominant regime, with controlled transitions to Hot state for active computation.

The sustained utilization factor is derived from the thermal duty-cycle analysis of Section III-G4:

$$\eta = 0.85 \tag{256}$$

This factor accounts for:

- JIR state-transition overhead ($\approx 4\ \mu$s per rotation cycle),
- PCM reconfiguration cycles (sparse, $\ll 1\%$ overhead under sustained inference),
- CRT reconstruction pipeline flushing (contained within the $\tau_{\text{JIR}} = 5\ \mu$s cycle, introducing no per-operation stall).

a) Simultaneous Weight Loading and Zero-Overhead Reconfiguration.: PCM reconfiguration introduces far less than 1% overhead because the JANUS architecture reprograms *all* switches simultaneously rather than sequentially. The entire weight fabric — up to 1,006,632,960 PCM switches in the Datacenter configuration — could, in principle, be reprogrammed in a single **0.1 ms** window by exploiting the Z-axis vertical GaP optical pump beam.

During the weight-loading window, the same GaP pump source ordinarily used for PSA parametric gain is repurposed as the PCM write beam. JANUS employs 32 dedicated optical pump lasers, one positioned below each tile, rather than a single centralized source. Each tile laser delivers focused write pulses to its local PCM switch array, simultaneously addressing all switches within that tile in parallel. This 32-laser arrangement could provide spatial selectivity across tiles while delivering the required 10–100 pJ write energy to each PCM switch. The distributed pump architecture also avoids GaP pump depletion effects that would arise from a single laser illuminating all 32 tiles simultaneously. This elegant dual-use of the GaP pump PSA gain during computation, PCM write during reconfiguration, in principle, eliminates the need for any dedicated per-switch electrical programming circuitry.

Furthermore, the 0.1 ms weight-loading window is not idle for the CMOS backend. The accumulated CRT reconstruction backlog from the previous compute epoch is processed during this interval. Because the CMOS pipeline is deeply staged, the reconstruction of the final batch of optical results from the previous epoch overlaps entirely with the optical write operation, yielding **zero effective CMOS stall overhead** attributable to reconfiguration. The net system utilization therefore remains at $\eta = 0.85$ rather than degrading with model-weight update frequency.

b) Reconfiguration Energy Budget.: PCM state transitions require write energy E_{PCM} per switch, which varies with device geometry, active volume, and the crystallisation (SET) versus amorphisation (RESET) direction. For plasmonic PCM switches of the type employed in JANUS, three design tiers bound the operating envelope across $N_{\text{sw}} = 1{,}006{,}632{,}960$ switches:

- **Tier 1 (highly optimised, $\approx$100 fJ):** Sub-wavelength MIM plasmonic structures, primarily crystallisation pulses.

$$E^{\text{T1}}_{\text{reconfig}} = 10^9 \times 100 \times 10^{-15} = 100\,\mu\text{J}, \tag{257}$$

$$P^{\text{T1}}_{\text{transient}} = \frac{100\,\mu\text{J}}{0.1\,\text{ms}} = 1\,\text{W}. \tag{258}$$

- **Tier 2 (balanced neuromorphic, $\approx$1 pJ):** Mixed SET/RESET operations across a standard plasmonic array.

$$E^{\text{T2}}_{\text{reconfig}} = 10^9 \times 1 \times 10^{-12} = 1\,\text{mJ}, \tag{259}$$

$$P^{\text{T2}}_{\text{transient}} = \frac{1\,\text{mJ}}{0.1\,\text{ms}} = 10\,\text{W}. \tag{260}$$

- **Tier 3 (conservative upper bound, $\approx$10 pJ):** Larger active volumes, unoptimised thermal leakage, or consecutive high-energy RESET pulses.

$$E^{\text{T3}}_{\text{reconfig}} = 10^9 \times 10 \times 10^{-12} = 10\,\text{mJ}, \tag{261}$$

$$P^{\text{T3}}_{\text{transient}} = \frac{10\,\text{mJ}}{0.1\,\text{ms}} = 100\,\text{W}. \tag{262}$$

Even under the most conservative Tier 3 scenario, the total transient power dissipation across the entire chip reaches only 100 W during the 0.1 ms write window. Distributing this across the 300 mm^2 Datacenter die through 32 spatially isolated tile pump lasers yields a peak transient power density of $\approx$0.33 W/mm^2, which is well within the thermal extraction capability of the External Thermal Cage and remains far below the $\sim$1 W/mm^2 limit demonstrated in high-performance silicon flip-chip packages. After the write window closes, the PCM state is latched non-volatilely and the transient power returns to zero; no sustained thermal load is imposed on the optical stack. In the operationally dominant mode — *incremental* reconfiguration updating only the fraction f_Δ of changed weights between inference epochs — the energy scales as $E_\Delta = f_\Delta \times E_{\text{reconfig}}$. For typical sparse fine-tuning updates ($f_\Delta < 1\%$), Tier 3 reduces to < 0.1 mJ per update, making reconfiguration energy negligible relative to inference throughput. Full simultaneous weight loading across all switches is therefore characterised as an "in-principle" upper bound; operational reprogramming exploits incremental sparsity via the 32 distributed tile pump lasers.

4) System Effective MAC Rate at Native 8-bit Precision: With N_{tiles} tiles operating in parallel at utilization η, the total system effective MAC rate at native 4-bit precision is:

$$\dot{M}_{\text{eff}} = N_{\text{tiles}} \times \dot{M}_{\text{tile}} \times \eta \tag{263}$$

Thus:

$$\dot{M}_{\text{Mini}} = 32 \times 33.8 \times 0.85 \approx 919.36 \text{ TMAC/s} \tag{264}$$

$$\dot{M}_{\text{Edge}} = 32 \times 135.2 \times 0.85 \approx 3{,}677 \text{ TMAC/s} \tag{265}$$

$$\dot{M}_{\text{Datacenter}} = 32 \times 540.7 \times 0.85 \approx 14{,}707 \text{ TMAC/s} \tag{266}$$

5) Precision Scaling via Spatial RNS: Higher-precision computation requires k tiles to compute independent RNS residues simultaneously, where k is determined by the target precision and the native per-tile precision of 8-bit. Here P denotes the bit-width of the multiplication *input operands*. The resulting product is $2P$ bits wide; $k = \lceil 2P/8 \rceil = \lceil P/4 \rceil$ optical tiles are required for exact product storage:

$$k = \left\lceil \frac{P}{4} \right\rceil = \left\lceil \frac{2P}{8} \right\rceil \tag{267}$$

Thus:

$$k_{\text{INT4 } (P=4)} : k = 1, \quad k_{\text{INT8 } (P=8)} : k = 2, \quad k_{\text{INT64 } (P=64)} : k = 16 \tag{268}$$

The effective precision-scaled throughput becomes:

$$\dot{M}_{\text{eff}}(P) = \frac{N_{\text{tiles}} \times \dot{M}_{\text{tile}} \times \eta}{k} \tag{269}$$

This linear inverse scaling in k is the direct cost of achieving exact higher precision through spatial modular composition. No analog signal quality is sacrificed; the trade-off is purely in tile allocation.

6) Energy Efficiency at 64-bit Precision: The energy efficiency at 64-bit exact operation is:

$$\eta_{\text{EE}} = \frac{\dot{M}_{\text{eff}}(\text{64-bit})}{P_{\text{total}}} \tag{270}$$

For the three configurations:

$$\eta_{\text{EE,Mini}} = \frac{57.6 \text{ TMAC/s}}{10 - 20 \text{ W}} \approx 5.76 - 2.88 \text{ TMAC/s/W} \tag{271}$$

$$\eta_{\text{EE,Edge}} = \frac{229.8 \text{ TMAC/s}}{50 - 60 \text{ W}} \approx 4.5 - 3.83 \text{ TMAC/s/W} \tag{272}$$

$$\eta_{\text{EE,DC}} = \frac{919.2 \text{ TMAC/s}}{120 - 140 \text{ W}} \approx 7.66 - 6.56 \text{ TMAC/s/W} \tag{273}$$

7) Sustained Throughput Summary Table: Table consolidates the complete throughput scaling across all three JANUS configurations and all target precisions. All values represent exact integer operations. No analog approximation is involved.

TABLE XVIII
JANUS MINI CONFIGURATION PERFORMANCE PROFILE (32 TILES, 32×32). NOTE: 1 MAC = 2 FLOPS (ONE MULTIPLY + ONE ADD); ALL THROUGHPUT FIGURES ARE MAC/S.

Precision Target	**RNS Tiles (k)**	**Throughput (TMAC/s)**	**EE (TMAC/s/W)**
8-bit Native	2	460.8	46.00 - 23.04
16-bit Exact	4	230.4	23.04 - 11.52
32-bit Exact	8	115.2	11.52 - 05.76
64-bit Exact	16	57.6	05.76 - 02.88

TABLE XIX
JANUS EDGE CONFIGURATION PERFORMANCE PROFILE (32 TILES, 64×64)

Precision Target	**RNS Tiles (k)**	**Throughput (TMAC/s)**	**EE (TMAC/s/W)**
8-bit Native	2	1,838.3	36.76 - 30.64
16-bit Exact	4	919.2	18.38 - 15.32
32-bit Exact	8	459.6	09.19 - 07.66
64-bit Exact	16	229.8	04.59 - 03.83

TABLE XX
JANUS DATACENTER CONFIGURATION PERFORMANCE PROFILE (32 TILES, 128×128)

Precision Target	**RNS Tiles (k)**	**Throughput (TMAC/s)**	**EE (TMAC/s/W)**
8-bit Native	2	7,353.2	61.27 - 52.52
16-bit Exact	4	3,676.6	30.63 - 26.26
32-bit Exact	8	1,838.2	15.31 - 13.13
64-bit Exact	16	919.1	07.65 - 06.56

K. Asymptotic Architectural Implications

The scaling behavior of JANUS differs fundamentally from conventional analog photonic accelerators.

Conventional architectures exhibit:

$$\text{Higher Scale} \rightarrow \text{Higher Instability} \tag{274}$$

because thermal drift, nonlinear interaction, calibration overhead, analog precision requirements, and fault observability all worsen with growth.

JANUS instead enforces bounded local operating conditions while scaling globally through modular spatial composition.

Thus:

$$\text{Higher Scale} \rightarrow \text{Higher Parallelism} \tag{275}$$

without proportionally increasing local physical instability.

This bounded-local-state principle is the primary architectural reason why JANUS remains physically scalable under exact arithmetic workloads.

V. PHYSICAL FEASIBILITY AND CONSTRAINT COMPLIANCE

A. Constraint-First Validation Philosophy

The validity of JANUS does not emerge from peak theoretical throughput claims alone. Instead, feasibility is determined by whether every subsystem remains consistent with known thermodynamic, optical, informational, and electrical constraints.

Therefore, each architectural mechanism must satisfy four conditions simultaneously:

- ☐ Physical Realizability
- ☐ Thermodynamic Stability
- ☐ Bounded Error Propagation
- ☐ Sustained Operational Equilibrium

Any subsystem violating one of these conditions becomes physically non-scalable regardless of theoretical arithmetic density.

Accordingly, JANUS was derived through iterative elimination of architectures that failed under one or more physical constraints.

The resulting system therefore represents a constraint-satisfying architecture rather than a performance-maximized approximation.

B. Shannon Capacity Compliance

The primary failure mode of conventional analog photonic accelerators is violation of Shannon-limited information recoverability.

For a communication channel with bandwidth: B and signal-to-noise ratio SNR the maximum recoverable information rate is:

$$C = B\log_2(1 + \mathrm{SNR}) \tag{276}$$

Conventional analog optical accumulation attempts to reconstruct large integer values directly from optical amplitude.

As previously derived, exact accumulation requires effective analog SNR approaching:

$$\mathrm{SNR}_{\mathrm{Mini}} \approx 126.4 \text{ dB} \tag{277}$$

$$\mathrm{SNR}_{\mathrm{Edge}} \approx 132.4 \text{ dB} \tag{278}$$

$$\mathrm{SNR}_{\mathrm{Datacenter}} \approx 138.4 \text{ dB} \tag{279}$$

Such requirements exceed realistic detector capability at 33 GHz operating frequencies.

JANUS resolves this contradiction by reformulating arithmetic into binary spatial localization.

The detector condition reduces to:

$$P_{\mathrm{signal}} > P_{\mathrm{noise}} \tag{280}$$

which corresponds approximately to:

$$\mathrm{SNR} > 0 \text{ dB} \tag{281}$$

Thus, JANUS remains consistent with Shannon information bounds because it no longer attempts exponentially precise analog amplitude recovery.

Instead, information is encoded spatially through deterministic One-Hot routing.

TABLE XXI
SHANNON COMPLIANCE COMPARISON BETWEEN ANALOG AND JANUS ARCHITECTURES

Metric	**Mini**	**Edge**	**Datacenter**
Analog Exact SNR Requirement	126.4 dB	132.4 dB	138.4 dB
JANUS Detection Requirement	> 0 dB	> 0 dB	> 0 dB
Detection Type	Binary	Binary	Binary
High-Resolution ADC Requirement	Eliminated	Eliminated	Eliminated
Shannon Feasibility	Bounded	Bounded	Bounded

C. Thermodynamic Compliance

Large-scale photonic systems commonly violate thermodynamic feasibility through excessive static routing power density.

For thermo-optic MZI routing:

$$P_{\mathrm{hold}} = N_{\mathrm{switch}} P_{\mathrm{MZI}} \tag{282}$$

with:

$$P_{\mathrm{MZI}} \approx 10 \text{ mW} \tag{283}$$

The resulting static thermal loads become:

$$P_{\mathrm{Mini}} = 314.57 \text{ kW} \tag{284}$$

$$P_{\mathrm{Edge}} = 2516.58 \text{ kW} \tag{285}$$

$$P_{\mathrm{Datacenter}} = 10{,}066.32 \text{ kW} \tag{286}$$

Such heat densities exceed realistic chip-scale thermal extraction limits.

JANUS eliminates this failure mechanism through non-volatile PCM switches.

After programming:

$$P_{\mathrm{hold}} \approx 0 \text{ W} \tag{287}$$

Thus, routing power scales with reconfiguration frequency rather than switch count.

Additionally, thermal isolation is enforced through directional Z-axis heat partitioning:

$$\frac{\partial T_{\mathrm{SiPh}}}{\partial T_{\mathrm{CMOS}}} \to 0 \tag{288}$$

This prevents optical routing drift caused by CMOS thermal activity.

Consequently, JANUS remains thermodynamically bounded under sustained operation.

D. Nonlinear Optical Compliance

Conventional DWDM photonic systems violate nonlinear optical stability at high channel density due to Kerr-induced spectral mixing.

The nonlinear refractive response becomes:

$$n = n_0 + n_2 I \tag{289}$$

which produces Four-Wave Mixing:

$$\omega_{FWM} = \omega_1 + \omega_2 - \omega_3 \tag{290}$$

The resulting nonlinear interference scales approximately as:

$$P_{\text{NL}} \propto \gamma P^3 L^2 \tag{291}$$

Thus, higher wavelength density eventually increases spectral corruption.

JANUS eliminates this failure mechanism structurally by enforcing:

$$N_\lambda = 1 \tag{292}$$

Without multiple carrier frequencies, inter-channel spectral mixing cannot occur.

Therefore:

$$P_{\text{ghost}} \to 0 \tag{293}$$

under bounded operating conditions.

This architecture remains consistent with nonlinear optical stability limits because nonlinear spectral interaction mechanisms are absent.

E. Amplification Noise Compliance

Conventional SOAs violate exact arithmetic recoverability because gain introduces amplified spontaneous emission noise:

$$\sigma^2_{\text{ASE}} \propto G n_{sp} B \tag{294}$$

Thus:

$$\text{SNR}_{out} < \text{SNR}_{in} \tag{295}$$

under conventional amplification.

JANUS instead employs degenerate phase-sensitive amplification with theoretical noise figure:

$$F_{\text{PSA}} \to 0 \text{ dB} \tag{296}$$

Distributed PSA gain satisfies:

$$G_{\text{PSA}} \approx L_{\text{routing}} \tag{297}$$

thereby maintaining approximately constant signal power throughout propagation:

$$P(z) \approx \text{constant} \tag{298}$$

Thus, JANUS satisfies bounded optical equilibrium conditions without catastrophic ASE accumulation.

F. Fault Detectability Compliance

Conventional approximate systems permit silent arithmetic corruption.

Under graceful degradation:

$$X \to X + \epsilon \tag{299}$$

small faults remain numerically plausible.

JANUS instead enforces arithmetic invalidation through CRT inconsistency amplification.

A residue fault:

$$x_j \to x_j + \delta \tag{300}$$

produces reconstructed deviation:

$$X' = X + \delta M_j N_j \mod M \tag{301}$$

Thus:

$$|X' - X| \gg |\delta| \tag{302}$$

and local faults become globally obvious.

The probability of silent undetected corruption therefore approaches:

$$P_{\text{silent}} \to 0 \tag{303}$$

under bounded fault assumptions.

This behavior satisfies deterministic exact arithmetic visibility requirements absent in conventional approximate architectures.

G. Calibration Stability Compliance

Large interferometric photonic meshes often require continuous dynamic calibration due to thermal phase drift:

$$\Delta\phi = \frac{2\pi L}{\lambda} \frac{dn}{dT} \Delta T \tag{304}$$

In conventional architectures:

$$H(t) \neq H(t + \Delta t) \tag{305}$$

Thus, routing correctness continuously evolves over time.

JANUS suppresses this instability through:

1) non-volatile PCM routing,
2) thermal Z-axis partitioning,
3) single-wavelength operation,

TABLE XXII
CONSTRAINT COMPLIANCE SUMMARY FOR JANUS

Constraint	**Conventional Failure**	**JANUS Resolution**
Shannon Precision Limit	Exponential SNR growth	One-Hot binary spatial localization
Thermodynamic Scaling	Kilowatt-scale routing heat	Zero-static-power PCM routing
Nonlinear Spectral Mixing	FWM and XPM corruption	Single-wavelength operation
Amplifier Noise	ASE accumulation	Near-zero-noise PSA amplification
Thermal Drift	Continuous calibration instability	Z-axis thermal decoupling
Silent Arithmetic Corruption	Graceful degradation	CRT arithmetic invalidation
Routing Crosstalk	Waveguide interference	Dilated Beneš separation
Precision Scaling	Analog precision escalation	Spatial modular composition

4) bounded optical equilibrium.

Consequently:

$$H(t) \approx \text{constant} \tag{306}$$

under sustained operating conditions.

Therefore, the architecture avoids the runaway calibration overhead commonly observed in large interferometric meshes.

H. Physical Constraint Summary

The architectural validity of JANUS emerges from simultaneous compliance with all dominant large-scale photonic constraints.

Importantly, each JANUS subsystem exists because a conventional alternative violates at least one physical law at large scale.

Thus, the architecture was not assembled through arbitrary feature selection.

It emerged through iterative elimination of physically unstable alternatives.

I. Feasibility Interpretation

JANUS does not claim violation of physical limits.

Instead, it explicitly reorganizes computation to remain inside them.

The architecture avoids:

- exponential analog precision scaling,
- unbounded thermal density,
- spectral nonlinear instability,
- unrecoverable amplification entropy,
- silent arithmetic degradation.

This distinction is critical.

Many photonic accelerator proposals attempt to overpower physical constraints through additional calibration, higher laser power, or more aggressive analog reconstruction.

JANUS instead reformulates the computational problem itself so that the constraints become structurally bounded.

Consequently, scalability emerges not from violating physical laws, but from remaining consistent with them.

VI. Discussion

A. Reframing the Objective of Photonic Computation

Most photonic accelerator research attempts to maximize arithmetic density by directly emulating conventional electronic matrix multiplication in the optical domain [3], [18], [36], [44].

This approach implicitly assumes that optical systems can scale analog precision similarly to digital electronics.

However, the preceding analysis demonstrates that this assumption fails under large-scale exact arithmetic workloads.

Thermal drift, optical shot noise, ADC precision limits, nonlinear spectral mixing, and amplifier entropy accumulation collectively impose hard physical ceilings on analog optical precision scaling.

Consequently, the central contribution of JANUS is not simply a faster photonic accelerator.

Instead, JANUS reformulates the computational paradigm itself.

The architecture abandons:

- analog amplitude accumulation,
- spectral multiplexing,
- volatile routing,
- continuous calibration,
- graceful degradation arithmetic.

and replaces them with:

- binary spatial localization,
- single-wavelength operation,
- non-volatile routing,
- bounded thermodynamic equilibrium,
- CRT-based arithmetic fault visibility.

Thus, JANUS should not be interpreted as an incremental extension of conventional analog photonics.

It represents a categorical shift from "analog optical approximation" toward "physically bounded exact spatial computation."

B. Why Exact Arithmetic Requires Spatial Reformulation

A key result of this work is that exact arithmetic and analog optical accumulation are fundamentally incompatible at large scale.

Conventional analog systems require exponentially increasing SNR:

$$\text{SNR} \propto 2^N \tag{307}$$

for exact reconstruction of large accumulations.

This growth rapidly exceeds detector, amplifier, and thermal stability limits.

JANUS resolves this contradiction by converting arithmetic precision into a spatial localization problem.

Instead of recovering amplitude magnitude:

$$\text{“How much light exists?”} \tag{308}$$

the detector solves:

$$\text{“Which waveguide contains light?”} \tag{309}$$

This distinction fundamentally changes the informational requirements of optical computation.

The resulting system no longer depends on exponentially precise analog recovery.

Consequently, exact arithmetic becomes physically compatible with high-frequency photonic operation.

C. Implications for Future Photonic Architectures

The architectural implications extend beyond JANUS itself.

Many existing photonic proposals rely on three assumptions:

1) high-resolution analog accumulation is scalable,
2) spectral multiplexing indefinitely increases bandwidth density,
3) continuous calibration can compensate arbitrary drift.

The present analysis suggests that all three assumptions become unstable at sufficiently large scale.

First, Shannon-limited information recovery prevents indefinite analog precision growth.

Second, nonlinear spectral interaction fundamentally limits DWDM scaling.

Third, calibration overhead eventually becomes thermodynamically and computationally dominant.

Therefore, future large-scale photonic systems may require:

- bounded local precision,
- spatial parallelism,
- non-volatile routing,
- thermally partitioned architectures,
- explicit fault visibility mechanisms.

These principles collectively define a constraint-bounded design methodology for scalable photonic computation.

D. On the Role of CMOS in Exact Optical Systems

JANUS intentionally preserves CMOS participation in the computational stack. This decision is critical. Many optical accelerator proposals attempt to eliminate electronic logic entirely in pursuit of all-optical computation. However, exact arithmetic verification, CRT reconstruction, fault visibility, memory coherence, and orchestration remain fundamentally digital tasks.

Thus, JANUS adopts a hybrid philosophy:

$$\text{Photonics} \rightarrow \text{Massively Parallel Transport} \tag{310}$$

$$\text{CMOS} \rightarrow \text{Deterministic Verification} \tag{311}$$

This division allows each physical substrate to operate within its native strengths. Photonics provides bandwidth density and passive propagation. CMOS provides deterministic state control and exact logical validation. The architecture therefore avoids forcing either substrate beyond its natural operating regime.

E. The Importance of CRT Arithmetic Visibility

A particularly important property of JANUS is the rejection of graceful degradation. Conventional approximate accelerators often prioritize throughput over deterministic correctness. As a result, transient faults may silently perturb numerical outputs. Such behavior becomes increasingly dangerous as systems scale. JANUS instead enforces arithmetic invalidation through CRT sensitivity. Small local residue faults become globally inconsistent reconstructed states. Consequently, the system physically refuses to generate numerically plausible incorrect outputs. This distinction has major implications for:

- scientific computing,
- cryptographic arithmetic,
- deterministic simulation,
- high-assurance inference,
- fault-sensitive computation.

In these domains, silent corruption is often more dangerous than total failure.

Thus, CRT arithmetic visibility represents a shift from “best-effort approximation” toward “explicit epistemic integrity.”

F. Scalability Through Bounded Local Complexity

Another major architectural principle emerging from JANUS is bounded local complexity.

Conventional scaling often increases local subsystem burden:

$$\text{Higher Scale} \rightarrow \text{Higher Local Precision} \tag{312}$$

or:

$$\text{Higher Scale} \rightarrow \text{Higher Thermal Density} \tag{313}$$

JANUS instead keeps local physical behavior approximately constant while increasing global computational parallelism.

Thus:

$$\text{Global Precision} \propto N_{\text{tiles}} \tag{314}$$

while:

$$P_{\text{local}} \approx \text{constant} \tag{315}$$

This bounded-local-state property is one of the primary reasons the architecture remains physically scalable.

Rather than overpowering local physical constraints, JANUS avoids creating them.

G. Limitations and Open Challenges

Although JANUS resolves several dominant scaling contradictions, three genuine open challenges remain at the system integration level.

First, extension of 3D SiPh stacking to five strata at JANUS switch densities. Three-dimensional silicon photonic integration with interlayer couplers and routing switches across multiple strata has been demonstrated using DUV stepper-compatible deposition processes, with two- and three-stratum platforms reported in recent literature [16]. However, extending this to the five-stratum configuration required by the JANUS datacenter fabric—accommodating 203.26 million PCM switches across a $\approx 251.7\ \mathrm{mm}^2$ per-layer chip footprint—has not yet been characterized for fabrication yield or interlayer optical loss accumulation across all five strata simultaneously.This represents the primary integration-scale challenge for the datacenter configuration.

Second, distributed degenerate GaP PSA pump phase-locking at scale. Phase-sensitive amplification in individual integrated waveguides has been experimentally demonstrated in silicon photonic crystal waveguides and chalcogenide platforms. GaP-on-insulator nonlinear photonics, including second-harmonic generation and frequency comb operation, has been demonstrated via direct wafer bonding and confirmed compatible with CMOS substrates. However, distributing a phase-coherent degenerate pump signal across millions of spatially separated GaP amplification sites—while maintaining the phase-matching conditions required for low-noise parametric gain throughout a large routing fabric—has not been demonstrated. Pump phase drift across the distributed network is the central unsolved problem, not the amplification mechanism itself.

Third, formal multi-fault analysis of CRT arithmetic visibility bounds. CRT magnitude amplification provides strong fault visibility under single-residue corruption. However, a formal analysis of detection probability under simultaneous multi-tile faults—including correlated failure modes such as a shared thermal event affecting adjacent residue channels—and an adversarial fault model covering faults confined within a single residue domain, remains an open theoretical problem. This analysis is required to bound the undetected error probability under worst-case operating conditions. The JIR thermal-prediction model additionally requires formal characterisation of the sensor update rate and temperature-trend threshold needed to guarantee RRNS engagement before a fault manifests.

Additionally, while CRT-based fault visibility suppresses silent numerical corruption, it does not eliminate the possibility of hard physical failure. It converts hidden numerical corruption into explicit detectable invalid states, enabling clean fault recovery rather than silent result poisoning.

H. Broader Interpretation

More broadly, JANUS suggests that future high-performance computing systems may increasingly depend upon:

$$\textit{constraint-aware computational reformulation} \tag{316}$$

rather than raw component scaling alone.

Historically, many computational advances emerged not from violating physical limits, but from reorganizing computation to avoid pathological operating regimes.

Examples include:

- packet switching replacing circuit exclusivity,
- cache hierarchies replacing uniform memory access,
- parallel vectorism replacing scalar frequency scaling.

JANUS follows a similar trajectory.

Instead of attempting infinitely precise analog photonic arithmetic, the architecture restructures computation into bounded spatially localized operations compatible with physical law.

Thus, the primary contribution of JANUS is not merely architectural.

It is methodological.

The system demonstrates that scalable photonic exact arithmetic may require redesigning the computational abstraction itself around the constraints imposed by thermodynamics, information theory, and nonlinear optics.

VII. Conclusion

This work introduced JANUS, a constraint-bounded photonic architecture for exact large-scale arithmetic computation.

Unlike conventional analog photonic accelerators, JANUS avoids high-precision optical accumulation entirely and instead reformulates computation around One-Hot spatial routing, residue decomposition, single-wavelength coherent transport, non-volatile PCM Beneš switching, distributed phase-sensitive amplification, and CRT-based exact reconstruction.

The architecture was derived through iterative elimination of physically unstable alternatives.

- First, analog accumulation was shown to violate Shannon-limited information recovery requirements at high precision.
- Second, thermo-optic routing was shown to become thermodynamically non-scalable under large routing densities.
- Third, conventional SOA amplification was shown to accumulate irrecoverable ASE entropy.
- Fourth, DWDM scaling was shown to induce nonlinear spectral corruption.
- Fifth, graceful degradation arithmetic was shown to permit silent numerical hallucination.

JANUS resolves these contradictions through structural architectural reformulation.

- Binary spatial localization replaces analog magnitude recovery.
- Single-wavelength coherent fanout removes inter-channel spectral mixing.
- PCM routing eliminates static hold power.
- Distributed PSA amplification maintains bounded optical equilibrium.

- CRT reconstruction converts local arithmetic faults into globally detectable invalid states.
- Finally, Z-axis thermal partitioning isolates the optical compute layer from dominant CMOS thermal activity.

Collectively, these mechanisms ensure that the architecture remains consistent with:

- Shannon information limits,
- thermodynamic stability,
- nonlinear optical bounds,
- bounded noise propagation,
- deterministic fault visibility.

Importantly, JANUS does not claim to transcend physical law.

Instead, it reorganizes computation so that all dominant physical constraints remain bounded simultaneously.

The resulting system demonstrates that scalable photonic exact arithmetic is achievable not through increasingly aggressive analog precision, but through spatial modular reformulation aligned with the underlying laws of optics, thermodynamics, and information theory.

Appendix A
Derivation of Analog Accumulation Precision Collapse

A. Maximum Exact Accumulation Requirement

Consider an analog photonic multiply-accumulate operation using unsigned 8-bit operands.

The maximum operand value becomes:

$$x_{\max} = 255 \tag{317}$$

The largest exact product is therefore:

$$p_{\max} = 255 \times 255 = 65025 \tag{318}$$

For an accumulation of:

$$N \tag{319}$$

terms, the maximum analog accumulation becomes:

$$S_{\max} = N \times 65025 \tag{320}$$

Thus:

$$S_{\text{Mini}} = 32 \times 65025 \tag{321}$$
$$= 2{,}080{,}800 \tag{322}$$

$$S_{\text{Edge}} = 64 \times 65025 \tag{323}$$
$$= 4{,}161{,}600 \tag{324}$$

$$S_{\text{Datacenter}} = 128 \times 65025 \tag{325}$$
$$= 8{,}323{,}200 \tag{326}$$

B. Equivalent ADC Resolution

The required exact ADC resolution becomes:

$$N_{\text{ADC}} = \lceil \log_2(S_{\max}) \rceil \tag{327}$$

Therefore:

$$N_{\text{ADC,Mini}} = \lceil \log_2(2{,}080{,}800) \rceil \tag{328}$$
$$= 21 \text{ bits} \tag{329}$$

$$N_{\text{ADC,Edge}} = \lceil \log_2(4{,}161{,}600) \rceil \tag{330}$$
$$= 22 \text{ bits} \tag{331}$$

$$N_{\text{ADC,Datacenter}} = \lceil \log_2(8{,}323{,}200) \rceil \tag{332}$$
$$= 23 \text{ bits} \tag{333}$$

Such resolutions are physically unrealistic at 33 GHz bandwidth.

C. Effective SNR Requirement

The theoretical SNR required for exact recovery of an N-bit converter becomes:

$$\text{SNR}_{\text{dB}} = 6.02N + 1.76 \tag{334}$$

Substituting the required ADC resolutions:

$$\text{SNR}_{\text{Mini}} = 6.02(21) + 1.76 \tag{335}$$
$$= 128.18 \text{ dB} \tag{336}$$

$$\text{SNR}_{\text{Edge}} = 6.02(22) + 1.76 \tag{337}$$
$$= 134.20 \text{ dB} \tag{338}$$

$$\text{SNR}_{\text{Datacenter}} = 6.02(23) + 1.76 \tag{339}$$
$$= 140.22 \text{ dB} \tag{340}$$

These values exceed practical photonic detector limits by several orders of magnitude.

TABLE XXIII
Analog Precision Collapse Derivation Summary

Metric	Mini	Edge	Datacenter
Accumulation Terms	32	64	128
Maximum Exact Sum	2,080,800	4,161,600	8,323,200
Required ADC Resolution	21-bit	22-bit	23-bit
Required SNR	128.18 dB	134.20 dB	140.22 dB
Physical Feasibility	No	No	No

Appendix B
Derivation of Beneš Routing Complexity

Note: The following derives the switch count for a *single per-multiplier Beneš fabric instance* operating on the 256-waveguide One-Hot alphabet. The system-total switch count equals this value $\times N^2$ multipliers per tile $\times N_{\text{tiles}}$. See Section II.C for the full system-total figures.

A. Stage Scaling

The JANUS Beneš fabric routes the fixed 256-waveguide One-Hot channel alphabet. A 256-port Beneš network requires:

$$S = 2\log_2(256) - 1 = \mathbf{15} \text{ stages (universal)} \tag{341}$$

This stage count is identical for all three deployment tiers because the routing granularity is the One-Hot channel count (256), not the matrix dimension N.

B. Switch Count Scaling

Each stage contains $256/2 = 128$ switching elements.

Therefore, the total switch count *per single Beneš fabric instance* is:

$$N_{\text{switch,single}} = \frac{256}{2}\left(2\log_2(256) - 1\right) = 128 \times 15 = 1{,}920 \tag{342}$$

Thus, for a single fabric instance (all tiers):

$$N_{\text{switch,single}} = 1{,}920 \tag{343}$$

The complete system-total switch counts (Section II.C) multiply this by the full number of per-multiplier Beneš fabrics deployed across all tiles.

C. Asymptotic Complexity

Asymptotically:

$$N_{\text{switch}} \in \mathcal{O}(N \log N) \tag{344}$$

which is substantially smaller than full crossbar complexity:

$$N_{\text{crossbar}} \in \mathcal{O}(N^2) \tag{345}$$

Therefore, Beneš routing remains scalable at large mesh sizes.

Appendix C
Derivation of Passive Fanout Loss

A. Ideal Power Splitting

For ideal passive fanout into:

$$N \tag{346}$$

branches:

$$P_k = \frac{P_0}{N} \tag{347}$$

The splitting loss becomes:

$$L_{\text{split}} = 10\log_{10}(N) \tag{348}$$

Thus:

$$\begin{aligned} L_{\text{Mini}} &= 10\log_{10}(32) &(349)\\ &= 15.05 \text{ dB} &(350)\end{aligned}$$

$$\begin{aligned} L_{\text{Edge}} &= 10\log_{10}(64) &(351)\\ &= 18.06 \text{ dB} &(352)\end{aligned}$$

$$\begin{aligned} L_{\text{Datacenter}} &= 10\log_{10}(128) &(353)\\ &= 21.07 \text{ dB} &(354)\end{aligned}$$

B. Implications

Because JANUS employs coherent single-wavelength fanout, all branches remain phase coherent:

$$\Delta\phi \approx 0 \tag{355}$$

Thus, passive splitting does not introduce spectral incoherence or inter-channel wavelength instability.

Appendix D
Derivation of Thermal Phase Drift

A. Temperature-Dependent Refractive Index

Silicon refractive index varies with temperature:

$$n(T) = n_0 + \frac{dn}{dT}\Delta T \tag{356}$$

Optical phase becomes:

$$\phi = \frac{2\pi n L}{\lambda} \tag{357}$$

Therefore:

$$\Delta\phi = \frac{2\pi L}{\lambda}\frac{dn}{dT}\Delta T \tag{358}$$

B. Implications

Even small thermal perturbations produce significant optical phase drift.

Thus, large-scale photonic routing requires bounded thermal environments.

JANUS achieves this through Z-axis thermal partitioning.

Appendix E
Derivation of CRT Error Amplification

A. CRT Reconstruction

For coprime modulus set:

$$\mathcal{M} = \{m_1, m_2, \ldots, m_k\} \tag{359}$$

the reconstructed integer becomes:

$$X = \sum_{i=1}^{k} x_i M_i N_i \mod M \tag{360}$$

where:

$$M = \prod_{i=1}^{k} m_i \tag{361}$$

B. Residue Fault

Suppose one residue becomes corrupted:

$$x_j \rightarrow x_j + \delta \tag{362}$$

Then reconstruction becomes:

$$X' = X + \delta M_j N_j \mod M \tag{363}$$

Thus:

$$|X' - X| \propto M_j \tag{364}$$

which scales with global CRT range.

Therefore, small local residue faults become globally amplified reconstruction inconsistencies.

C. Interpretation

The architecture therefore converts:

$$\text{Small Local Fault} \rightarrow \text{Large Global Invalidity} \tag{365}$$

This property forms the basis of CRT arithmetic fault visibility in JANUS.

Appendix F
Scope Boundaries and Open Problems

A. What JANUS Does Not Claim

Intellectual honesty requires explicit statement of the boundaries of this work. JANUS does not claim:

- **Perfect fault immunity.** CRT consistency checking detects fault patterns that produce residue inconsistency, but multi-fault combinations or faults confined within a single residue domain may remain undetected without redundant moduli. The JIR mitigates this through predictive RRNS engagement on thermal trend, but formal closed-form bounds remain an open problem.
- **Elimination of photonic nonlinearities.** Single-wavelength operation substantially suppresses inter-channel nonlinear interactions. Self-phase modulation and intra-channel effects remain present and must be managed through power budgeting.
- **Replacement of all digital arithmetic.** JANUS targets photonic matrix multiplication workloads. It is not proposed as a general-purpose processor replacement.
- **Fabrication-validated product-scale implementation.** The full architecture remains a forward-looking systems integration proposal. Component-level demonstrations from the literature motivate the design, but full system integration has not yet been demonstrated.
- **Analog noise elimination.** JANUS does not eliminate analog noise, fabrication variation, or thermal fluctuation. Instead, it reduces the system's dependence on global high-resolution analog reconstruction, trading analog precision requirements for spatial routing correctness.

B. Future Work and Open Problems

Three open problems must be resolved for JANUS to progress from architectural proposal to fabrication-validated implementation.

- **Five-stratum SiPh yield characterization.** Recent work has demonstrated CMOS-compatible 3D silicon photonic platforms with interlayer couplers and switching elements across two and three strata, fabricated using standard DUV stepper processes. The JANUS datacenter configuration requires extending this to five strata while maintaining acceptable interlayer optical coupling loss and fabrication yield across 7.86 million PCM switch sites within a single reticle field. The key open questions are cumulative interlayer insertion loss across five strata, and wafer-scale yield of vertical optical vias at the required density. These are questions of demonstrated-process extension, not of unproven technology.
- **Distributed degenerate PSA pump phase-locking.** GaP-on-insulator nonlinear photonics has been demonstrated via direct wafer bonding, and the process has been confirmed compatible with CMOS substrates [4]. Degenerate phase-sensitive amplification has been experimentally demonstrated in integrated waveguide platforms. The remaining open problem is maintaining phase coherence of the degenerate pump across a spatially

distributed network of GaP amplification zones spanning a multi-tile datacenter fabric. Pump phase drift across spatial separation is the specific unsolved engineering problem, requiring an on-chip phase distribution and locking architecture whose design and characterization remains future work.

The JANUS architecture partially mitigates this challenge through four complementary structural strategies that collectively reduce dependence on active electronic phase-locking circuits.

(i) GaP Layer Routing Isolation. All electrical routing, vias, and signal interconnects are directed entirely outside the GaP medium. By treating each GaP segment strictly as a passive, localized nonlinear crystal zone rather than an active routing tier, the architecture eliminates localized electromagnetic interference, metal-induced optical absorption, and current-injection gradients that would otherwise dynamically perturb the refractive index through thermo-optic (dn/dT) or carrier-plasma (dn/dN) effects. This isolation preserves the phase predictability of each amplification zone.

(ii) Segmented GaP Island Growth. Rather than bonding a continuous GaP layer across the full reticle field—which would introduce macroscopic lattice mismatch strain, wafer bowing, and elevated defect densities across the Silicon/SiO_2 interface—the JANUS architecture grows or bonds discrete GaP islands only at locations aligned with waveguide inverse tapers where amplification is strictly required. This segmented approach relieves mechanical strain, improves fabrication yield, and localizes any residual phase-sensitivity variation to individually bounded, low-area zones.

(iii) Spatial Phase Uniformity via Global Vertical Pump Illumination. JANUS employs a single vertically incident pump laser to illuminate the full 3D stack from above. Because the pump phase front arrives across the chip plane with uniform spatial phase, the relative phase condition for degenerate parametric gain,

$$\Delta\phi = \phi_{\text{pump}} - 2\phi_{\text{signal}}, \tag{366}$$

becomes determined entirely by the optical path length accumulated by the signal waveguide prior to entering each GaP island. This transfers the phase-matching burden from dynamic electrical compensation onto static geometric layout engineering: silicon routing waveguides leading to parallel residue tiles are designed with precisely matched optical path lengths, so that signals arrive at their respective GaP zones in-phase with the global pump by construction.

(iv) Exploitation of the $100\ \mu\text{m}$ *Physical Scale Limit.* Phase mismatch accumulation scales as $\Delta\beta \cdot L$, where L is the interaction length. Over the $100\ \mu\text{m}$ GaP interaction zones employed by JANUS, the available propagation length is insufficient to accumulate significant phase drift even under minor material non-uniformities. Furthermore, at this microscopic scale, thermal gradients across the GaP medium are negligibly small, suppressing dynamic refractive index perturbations through the thermo-optic effect. Together, these scale-dependent properties allow the phase velocities of signal and pump to remain substantially synchronized through native waveguide geometry rather than active feedback.

Collectively, these four strategies replace active phase-locking electronics with static physical architecture:

$$\begin{gathered} \textbf{Active Phase-Locking Circuits} \\ \downarrow \\ \begin{cases} \text{1. Passive GaP island isolation (no routing wires)} \\ \text{2. Segmented island growth (strain relief)} \\ \text{3. Single-}\lambda\text{ uniform 2D pump front} \\ \text{4. } 100\,\mu\text{m scale-limited interaction zones} \end{cases} \end{gathered} \tag{367}$$

Nonetheless, formal experimental characterization of residual pump phase coherence across a full multi-tile datacenter fabric, and quantification of the phase error budget attributable to inter-island path-length fabrication tolerances, remain open engineering problems for future prototype integration.

- **Third, formal multi-fault analysis of CRT arithmetic visibility bounds.** Under single-residue corruption, CRT magnitude amplification provides strong fault visibility. Formal analysis is needed to characterize detection probability under simultaneous multi-tile faults, correlated failure modes affecting spatially adjacent residue channels, and faults that remain confined within a single residue domain without crossing into CRT inconsistency. Additionally, the JIR thermal-prediction model requires formal characterization of the temperature-trend detection latency and the minimum sensor update rate needed to engage RRNS protection before a fault materialises. Establishing closed-form detectability bounds under an adversarial fault model would complete the reliability argument for the architecture.

C. Notation Summary

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TABLE XXIV
PRIMARY NOTATION USED THROUGHOUT THIS WORK

SymbolDefinition	
N	Matrix/mesh dimension
P	Optical power (context-dependent)
P_p	PSA pump power
L	Propagation or interaction length
M	CRT product range: $M = \prod m_i$
M_j	CRT partial product: M/m_j
m_i	i-th RNS modulus
x_i	Residue of X modulo m_i
N_i	CRT Garner coefficient: $M_i^{-1} \bmod m_i$
γ	Nonlinear optical coefficient
G	Amplifier gain
F	Amplifier noise figure
S	Number of Beneš routing stages
ϕ	Optical phase
α	Optical attenuation coefficient (or thermal diffusivity, by context)
T	Temperature or throughput (by context)
n	Refractive index
$\chi^{(2)}$	Second-order nonlinear susceptibility
ω	Angular optical frequency

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